Specification Surfaces for Incremental Predictive Performance: Estimation, Uncertainty, and Multi-Log Evaluation

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Abstract

What a recorded field is worth to a prediction is reported as one number, but it is a functional of choices the report rarely states: the baseline, the pipeline, the outcome predicted, the split, the moment the decision is taken, how completely the register behind the field is populated, and the metric. We define incremental predictive performance as a *specification surface* $V_s(f)$ over a design space the analyst declares, and give three objects for reporting one: an exact functional-ANOVA decomposition of its variance that separates analyst choices from the resampling a design contains; simultaneous max-t bands from a pipeline-refitting bootstrap, whose coverage we measure against a known answer and find short of nominal; and *resolution regions*, which say where a sign is determined. On 19 pairs from 13 public event logs, resampling carries a median 71.7% of the variance, and net of it which analyst choice leads is a property of the pair. Most specifications leave the sign undetermined; where the data determine one, admissible specifications sit a median 0.0510 AUC from the conventional report. In a case study on 45,455 bank incidents, a register's worth turns on the decision time and its sign on the target.

Keywords: specification surface, incremental predictive performance, event logs, simultaneous inference, sensitivity analysis, configuration management database

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Highlights

- A field's incremental predictive performance is a surface, not one number
- An exact decomposition separates analyst choices from resampling
- Simultaneous bands over the whole declared family, with coverage measured
- Resolved cells sit 0.0510 AUC from the one-number report
- A register's worth turns on the decision time and on which target is set

1. Introduction

A claim that a recorded field is worth something is a comparison. Its second arm — performance *without* the field — is a choice the analyst makes, and so are the learner that makes the comparison, the encoding it uses, the outcome it predicts, the way the data are split in time, whether the field is available at the moment the decision is taken, how completely the register behind the field is populated, the metric the difference is read in, and the operating point at which that metric is evaluated. Each is routinely left implicit, and the result is reported as one number.

This paper's position is not that such a number depends on those choices — that is known, and Section 2 says by whom — but that the dependence is large enough, on real decisions, that the number alone is not a report; that the right object is therefore a surface over a *declared* design space; and that a surface admits estimation and inference of its own. Two things follow that a reader may not expect, and both are stated here before anything is argued for. **A pooled specification surface overstates the case**: most of its variance here is resampling rather than analyst latitude, and the correction is exact (Section 4.5). And **the sign is the wrong thing to count**: where the data determine a sign, the conventional report is usually right about it and usually silent about the size, which is what Section 6.4 reports instead.

The estimand. Write a specification s for a complete assignment to every design axis, and

$$V_s(f) \;=\; m_\theta(A_s(D_{\text{train}}, B \cup \{f_q\}),\, D_{\text{test}}) \;-\; m_\theta(A_s(D_{\text{train}}, B),\, D_{\text{test}})\,, \quad (1)$$

where A_s is the fitting procedure (learner, encoding, hyperparameters, calibration), B the baseline feature set, f_q the field under a register quality mechanism q, m the metric and θ the operating point, and where $D_{\text{train}}, D_{\text{test}}$ are fixed by the split rule and the cohort. The *specification surface* is the map $s \mapsto V_s(f)$ over a design space $\mathcal{S}$ that the analyst declares and the reader can enumerate; Figure 1 draws one.

We say *incremental predictive performance* and not *value*: (1) measures how much better a model predicts, not what the field costs to acquire or maintain, whether acting on the prediction changes the outcome, or what an organisation would pay for either. Section 8 converts the increment into net benefit at a declared exchange rate, which is as close to economic value as this evidence gets.

Contributions.

1. **An estimand, a declared surface and an exact decomposition** (Table 1). The surface over a design space the reader can enumerate; three *denominators* distinguished and audited from the files, so that a claim's quantifier is a counted set rather than a phrase; and an exact functional-ANOVA decomposition of the surface's variance in which the split is an *error stratum*, so that resampling is separated from the choices an analyst makes (Section 4.5).
2. **Simultaneous inference whose coverage is measured, and the labels that rest on it.** Simultaneous max-t bands over the whole family a claim ranges over, from a pipeline-refitting bootstrap; *resolution regions* with a robustness index ρ under a declared minimum resolved share; and, in Section 9.4, the band's family-wise coverage measured against a known answer — which is short of nominal everywhere in this corpus's configuration, so the labels are reported as descriptive diagnostics and not as guarantees.
3. **A registered multi-log benchmark and a register at three decision times.** The surface is computed through one frozen pipeline on every log–target pair a pre-registered protocol admits — 19 of them, not only the ones where the answer resolves — and every corpus statistic is reported on all 19 and on the 8 where the register is a maintained one (Section S14.17). Section 7 then shows that a decision time fixes which cases are in scope, what is known about them and which fields are admissible at once, reconciles the two values the case study's log has been reported to give, and finds the difference is the *target*: over

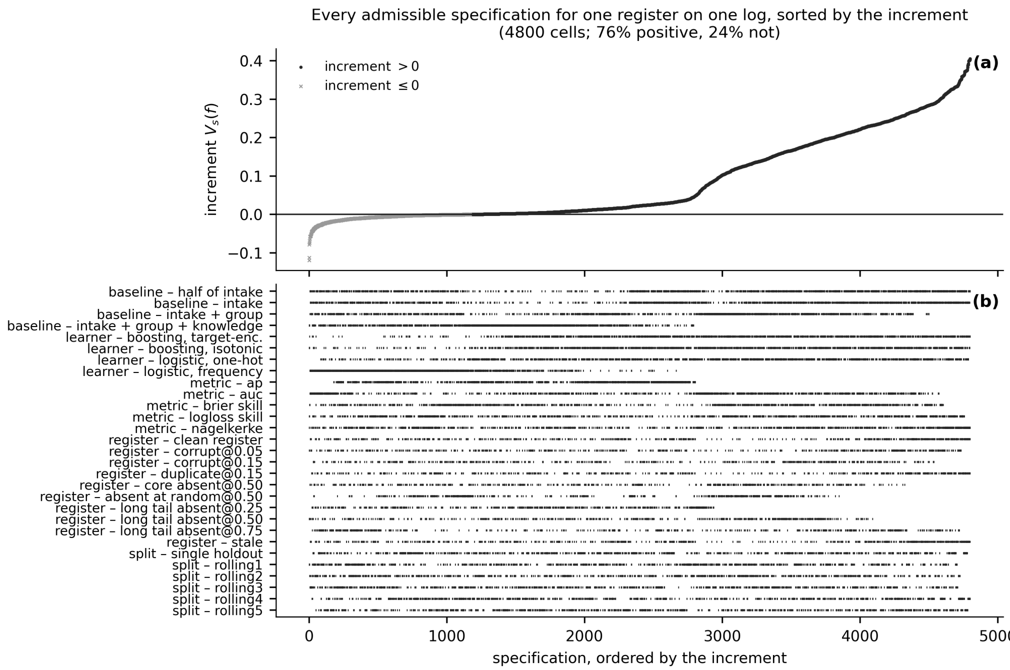

Figure 1: The specification curve for the configuration item on BPIC14, at the registered handover target: 4,800 admissible scalar cells, 4 pipelines × 4 baseline rungs × 5 instruments × 10 register-quality conditions × 6 splits. **(a)** Each point is one admissible specification — a complete assignment to those five axes — sorted by the increment. **(b)** Which level of which axis each specification occupies. The intercept-only rung is excluded. Every point is a point estimate; the simultaneous bands that decide the region labels are in Table S9. **The five scalar instruments are plotted on one axis and their increments are not on a common scale** (Section 4.7); what the panel supports is the *sign* statement, which is unit-free, and not a comparison of magnitudes across instruments. The panel shows the increment rather than the reduction, so a cell where the register is worth almost nothing looks the same as one where the free field absorbs almost everything.

the cohort × target design the target carries 99.3% of the variance and the cohort 0.0%.

Specification regret — what choosing one specification costs against the best, on a scale a decision can use — is reported in full and is *not* among the contributions above. Sections 6.6 and 8.4 report that the one-number rule's median excess regret is zero in every instrument, that the out-of-sample ordering reverses under rolling folds, and that the rules mostly tie in net benefit. It is carried as a *robustness check* on the other two arguments (Section 4.7):

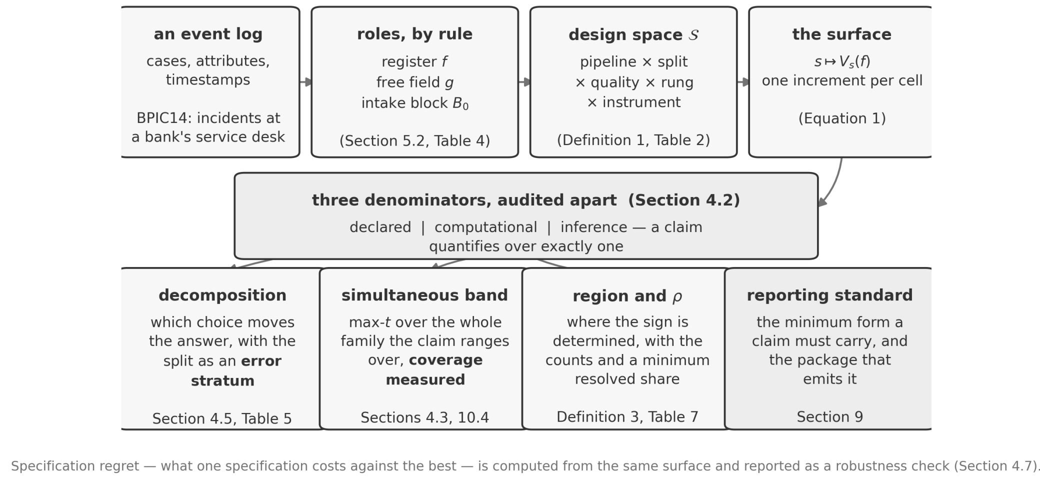

Figure 2: The method, once. A log is read into three *roles* by declared rules; the roles are crossed over a declared *design space*; the result is a *surface*, one increment per cell. Every claim about that surface names which of three *denominators* it quantifies over, and the reporting objects are computed from it. The schematic carries no result: the numbers are in the tables it names.

an object that reports a null is a check and not a result.

What this paper is not. This is a retrospective study on public benchmarks, offered as an evaluation methodology. It is not a procurement conclusion about configuration management databases, not a systematic review of reporting practice, and not a report of a deployed system; Section 10 says what each of those would have required.

The supplementary document. The protocol and its amendments, the proofs, the constructions of the reporting objects, the correction register, the results this article summarises and the tables it does not print are in a supplementary document whose sections, tables and figures carry an S. A reference here to Section S3 or Table S12 is a reference to that document. **The article states every claim it makes and prints the evidence for each**: which attribute plays the register on every pair and how many levels each axis carries (Table 3), the master table (Table 4), what a region label rests on (Table 6), the decomposition net of resampling (Table 5), the decision-time ladder (Table 8) and the band's measured coverage (Table S40). The supplement carries the constructions, the proofs and the sensitivity analyses; a reader who wants to check an inference rather than read one will want it,

object	what it is for	§	table	what it costs
Sensitivity decomposition	how much of the disagreement each analysis choice explains, and how much belongs to no single one	4.5	S8	exact; no refit. A Möbius sum over the surface already computed
Simultaneous band	an interval that holds over every cell a claim quantifies over, not one cell at a time	4.3	S9	the whole cost: one draw refits every arm of every cell
Resolution region and ρ	which way the surface points where the data settle it, and how much of it they settle	4.6	6	a function of the band; no further refit
Specification regret *(a check, not an object)*	what choosing one specification costs against the best one, on a scale a decision can use — it returns a null on this corpus	4.7	S18	exact; no refit, on the surface already computed

Table 1: THE REPORTING OBJECTS. The three this paper supplies and the fourth it retains as a robustness check, what each is for, where it is defined, where it is reported on this corpus, and what it costs; the two middle columns are the section it is defined in and the table that reports it. Only the second costs model fits: a draw of the bootstrap refits every arm of every cell, which is why the surface an inference is taken over is smaller than the surface a point estimate is taken over (Section 4.2). The other three are functions of a surface that has already been computed, so an analyst who has done the sweep has already paid for them.

and Supplement S10 is a glossary of every term used here in a technical sense, with the case study's own value for each.

2. Related work

Every component used here is established. That a feature's apparent importance depends on what else is in the model motivates conditional and permutation importance [40, 14], Shapley attributions [24, 8] and nonparametric formulations of importance as a population quantity [50]; the clinical-prediction literature has said for two decades that the increment in AUC from adding a predictor depends on the existing model [29, 7], is a poor guide to clinical usefulness [7, 45] and is routinely misread [21]; the same argument is made about effort-aware metrics in defect prediction [4], where which metric is chosen changes which method wins. Decision-analytic evaluation supplies the exchange rate [47, 48, 46]. Multiverse and specification-curve analysis supply the sweep [38, 36, 27], underspecification and variance studies the observation that pipelines differ [9, 2], and data valuation a different question about a datum's worth [15, 18]. On event logs, outcome-oriented predictive monitoring supplies the task [42, 44, 34], encoding studies one axis [23], and data-quality work the vocabulary for a degraded register [41, 26, 6]. Leakage discipline is standard [20, 19, 49].

Four further strands bear on the argument and are set out in Supplement S1.1 with what this paper takes from each: *many-analysts studies* [35, 3], which establish that analytic variability is large without supplying an estimand for it; *predictive multiplicity* and Rashomon sets [25, 33], which are multiplicity at the level of a fitted model where this paper's is at the level of the design that produced it; *benchmark-variance work in predictive process monitoring* [30], which is the evidence this paper's decomposition partitions; and the *sensitivity-analysis* tradition of bounding an unmeasured assumption [43], which is the form Section 7.3's leakage tipping point takes.

The novelty gap, precisely. Every component used here is established and none is claimed. What none of them supplies is the object this paper reports, and the gap is in four places. *The estimand is not declared completely*: the axes are discussed one at a time, and the decision time and the register's quality as preliminaries rather than as arguments. *The measure is left implicit*: a specification curve is summarised uniformly over whatever cells were enumerated, a choice that moves the answer (Section 4.5). *Inference is pointwise or absent*: a curve is a distribution of point estimates and a permutation test on its median, which licenses no statement about a sub-region. *Nothing measures what the conventional report costs*: the alternative to one number is offered as a display, not as a decision rule with a loss.

What is new here is narrower than the objects' names suggest, and is stated precisely. The decomposition is an adaptation of functional ANOVA over a configuration space [17]. *Counting the share of specifications with an effect in a declared direction is already standard specification-curve practice* [36]: the robustness index ρ is that count, and this paper does not claim it. What is new is what the count is taken over and under — a *simultaneous* band over a declared family rather than a pointwise test, an *audited* denominator rather than the cells that happened to be enumerated, and a minimum resolved share below which no direction is reported at all. The decomposition with the split as an error stratum, and the separation of analyst latitude from resampling that it makes exact, we have not seen elsewhere. Section 6.5 measures where a curve's verdict and a region label part company rather than asserting a difference.

The contribution is deliberately *not* "people usually report one number". It is that a claim of this form is a functional of a design, that the functional can be estimated and its uncertainty quantified, and that on this corpus it is far from constant.

3. The specification surface

3.1. The design space

Definition 1 (Design space)**.** *A* design space $\mathcal{S}$ *for a feature-value claim is a finite set of specifications, each a complete assignment to the axes*

$$s = \Big(\underbrace{A}_{\textit{learner}}, \underbrace{E}_{\textit{encoding}}, \underbrace{Y}_{\textit{target}}, \underbrace{\Pi}_{\textit{split}}, \underbrace{\tau}_{\textit{decision time}}, \underbrace{q}_{\textit{register quality}}, \underbrace{B}_{\textit{baseline}}, \underbrace{m}_{\textit{metric}}, \underbrace{\theta}_{\textit{operating point}} \Big),$$

together with the cohort and eligibility rule that fix D. A claim about a feature is well posed *only relative to a declared $\mathcal{S}$.*

Declaring a design space means writing down every axis, its levels, the measure over those levels, and every level deliberately excluded with the reason. Table 2 is that declaration, generated from the code that walks the space. **Not every axis is a factor of the cell count, and the table's 'in the product' column says which are.** The admissible cell counts of Section 6 are the product of the rows the table marks YES — the learner, the split, the register-quality condition, the admissible baseline and the scalar instrument — and that multiplication, re-done pair by pair from the same file the table is generated from, is a condition the verification harness checks. The rows marked NO are not omissions and are not free to be multiplied in: the encoding is fixed by the learner, so the learner row already counts the pipeline and multiplying the two would count it twice; the ladder-rung row is the admissible row plus the intercept-only baseline, which is in the surface

axis	levels	in the product	measure	exclusions
learner	2-4	yes	equal / ref. / conc.	none
encoding	2-3	no: in the learner	set by the learner	none
split	6	yes	equal / ref. / conc.	none
decision time (availability)	1-2	no: second surface	never averaged	two times on the primary log, one elsewhere
register quality x severity	6-10	yes	equal / ref. / conc.	none
baseline (ladder rung)	4-5	no: the row below	equal / ref. / conc.	intercept-only: in the surface, out of every admissible set
baseline, admissible only	3-4	yes	equal / ref. / conc.	none
scalar instrument	5	yes	never averaged across	none
operating point (net benefit)	31	no: decision curve	declared threshold dist.	outside the declared range

Table 2: The declared design space. One row per axis: the number of levels (a range where it differs by log), whether the axis is a FACTOR OF THE ADMISSIBLE CELL COUNT, the measure the paper places on those levels, and every level excluded by declaration with its reason. Generated from the code that walks the space. NOT EVERY ROW MULTIPLIES: only the rows marked 'yes' do, and their product is learner $\times$ split $\times$ register quality $\times$ admissible baseline $\times$ scalar instrument, which is the 'learners', 'splits', 'quality', 'rungs' and 'metrics' columns of Table S14 and equals its 'declared scalar' column on every pair — the multiplication the verification harness re-does, pair by pair, from the same file this table is generated from. The rows marked 'no' are not axes the study leaves unvaried; each is counted somewhere else, and the column says where. The encoding is determined by the learner — each learner names its own encoder — so the learner row already counts the pipeline and multiplying by the encoding would count it twice; Table S7 separates the two where they can be separated. The decision time indexes a SECOND surface on the case-study log rather than a further factor of the first, and Section 7.2 reports it. The ladder-rung row is the admissible row plus the intercept-only baseline, so the two rows are one axis at two admissibility conventions and only the admissible one enters an admissible count. And the operating point multiplies the decision-curve surface, not the scalar one, which is why the two surfaces are counted separately in Section 6.

and in no admissible set; the decision time indexes a second surface on the case study's log rather than a further factor of the first, and Section 7.2 reports it; and the operating point multiplies the decision-curve grid, which Section 6 counts separately from the scalar cells. **The nine axes above are the estimand's arguments, and this study does not vary all of them.** The target Y indexes the surfaces rather than being an axis *of* one, and the decision time, the operating point and the cohort vary on the case study's log alone; Table 3 states per pair which axes ran and at how many levels, and Section 10 carries the gap between the two tables. The *measure* column names which of the three declared product measures a quantity is taken under — EQUAL, REF. and CONC., placing equal, half and four fifths of the mass on the reference level — and Section 4.5 reports every decomposition under all three. The measure is a modelling choice, declared rather than left implicit; Supplement S1.2 states it, and says why the decision time and the register's quality are axes of the estimand rather than preliminaries to it.

3.2. What the surface estimates

Equation (1) is written on a fixed D_{train} and D_{test}, so as it stands $V_s(f)$ is a deterministic functional of the log in hand and an interval for it would cover with probability zero or one. That is not the quantity this paper reports intervals for, and the distinction has to be made before Section 4 can say what its bootstrap targets.

Definition 2 (The estimand)**.** *Let the cases of a log be a sample from a population P, and let the specification s fix the fitting procedure A_s, the cohort and eligibility rule, the split rule Π, the decision time, the register-quality mechanism, the baseline and the instrument. Write $A_s(P)$ for the limit of the fitted scoring rule as the sample grows under P with s held fixed.* ***That limit needs an assumption, and it is stated here rather than where it is used****: the case sequence is stationary and mixing, so that a temporal split does not separate two different populations and the moving-block bootstrap of Section 4.1 resamples a dependence structure that persists. An event log is not guaranteed to have that property and this paper does not test it; Section 10 carries it as a threat. The* estimand *at s is*

$$\vartheta_s(f) \;=\; m_\theta\Big(A_s(P, B \cup \{f_q\}),\, P\Big) \;-\; m_\theta\Big(A_s(P, B),\, P\Big),$$

and $V_s(f)$ of (1) *is its plug-in estimate on the log.*

Three things follow and are used throughout. The split rule and the pipeline are *part of the estimand*, not nuisances to be averaged away: ϑ_s is what this analyst's procedure converges to, which is why the nested bootstrap of Section 4.1 resamples the training half rather than conditioning on it, and why the split is an axis. Under misspecification ϑ_s is not what the register is worth in the world — the simulation of Section 9 separates the two and measures the gap, calling them V_{limit} and V_{oracle}. And every cell of every figure and table in this paper is an *estimate* of ϑ_s; "resolved" in Definition 3 means that the band excludes zero for ϑ_s, not that the observed increment is nonzero.

3.3. The surface, and why it is not a scalar

Given $\mathcal{S}$ and data D, the surface is the map $s \mapsto V_s(f)$ of (1), and a conventional report is one cell of it. The paper's claim is that a single cell is not a summary of the surface in any useful sense. **The point estimates alone establish that**: on 18 of the 19 pairs, between a tenth and nine tenths of the admissible cells are positive, so the surface's sign is not settled by the data and the design together. Whether that unsettledness amounts to a *misstatement* is a different and much harder question, and Section 6.4 answers it with three restrictions and a withdrawal rather than with a rate. There is one state in which a scalar is defensible: when the whole admissible surface lies on the same side of zero with simultaneous confidence, a positive number is a lower bound on a robust conclusion. Section 4.6 makes that a definition.

3.4. The relative reduction, and its pathologies

Comparisons between two baselines are often stated as a proportion. Write $B_0 \subset B_1$ and

$$D_s(f) \;=\; V_s(f \mid B_0) - V_s(f \mid B_1), \quad R_s(f) \;=\; 1 - \frac{V_s(f \mid B_1)}{V_s(f \mid B_0)} \;=\; \frac{D_s(f)}{V_s(f \mid B_0)}. \tag{2}$$

D is an absolute difference in the metric's own units. R is a ratio and inherits every pathology of one: the denominator can be small, zero or negative; R is not comparable across metrics, because Remark 1 says the two are not functions of each other; and R can exceed 1 or fall below 0, so reading it as "the share absorbed" is a category error whenever it leaves $[0, 1]$.

We therefore make D and the two increments primary throughout and treat R as a secondary descriptive statistic subject to three rules, declared before any R in this paper was computed:

1. R is reported only where $V_s(f \mid B_0)$ is resolvably positive under the *simultaneous* band, not merely positive as a point estimate;
2. its interval is a Fieller set and its *kind* — bounded, unbounded, or the complement of an interval — is printed beside it. A confidence set for a ratio is the solution set of a quadratic [13] and is a bounded interval only when that quadratic's leading coefficient is positive, which is to say only when the denominator is resolvably away from zero; otherwise it is unbounded or is the *complement* of an interval, and no bounded interval summarises it (Supplement S3). Of the 10,584 reductions this paper computes, 5,282 have a non-bounded Fieller set, and a percentile interval reports a bounded one for every single one of them;
3. the magnitude of R is never compared with the spread of R across metrics as though both lived on an unproblematic common scale.

3.5. Two facts about the metric, and why they are remarks

Two properties of the reduction are used later. The first is used in this article and is stated here; the second is used where it is proved, and is stated there. Supplement S2 carries both in full, their constructions, the richness assumption the second needs, and the witness that separates it from rank-basedness.

Remark 1 (The metric is an argument of the estimand)**.** *No function carries the AUC reduction to the average-precision reduction: two configurations can share an AUC reduction to* 2.2×10^{-16} *while their average-precision reductions differ by 0.417. The reduction is therefore not identified by the data, the feature and the two baselines, which is why every table in this paper names its instrument and why the minimal practically important difference of Section 6.4 is declared in one instrument's units and applied there only.*

The second — **Remark S1**, that a reduction is invariant to a declared family of recalibrations exactly when the metric is *affine* under it, so that one violating map rules the invariance out everywhere — is stated and proved in Supplement S2, with the 18 certificates it licenses.

4. Estimation and inference

4.1. The estimator, and uncertainty that includes the model

The pipeline is frozen and identical on every log, and the *pipeline* axis spans encodings as well as model families. Every learner, encoder and calibrator is the `scikit-learn` implementation [28] at the version `requirements.lock` pins, so the pipeline axis varies what this paper declares and nothing underneath it. Attributes are assigned to roles by the pre-registered rules of Section 5, cases are ordered by their first timestamp, and the split is temporal; Supplement S3.2 lists the levels and why the tree learner is target-encoded. The usual interval for such an increment resamples test rows with the models held fixed, which excludes training-sample, model-selection, encoder and split variability and the temporal dependence between cases.

Every interval here is instead a *nested moving-block bootstrap*: one draw resamples the training half by moving blocks [22] of length $\lceil n^{1/3} \rceil$, refits every arm of every rung under every learner and quality mechanism, and evaluates on a moving-block resample of the test half, with encoders, frequency tables, target encodings, register rankings and calibrators rebuilt inside the draw. On the simulation of Section 9 the fixed-model interval covers a median 92.1% and the nested basic interval 93.0% — **a gap of about 1.3 Monte Carlo standard errors** at the 0.7% standard error that design attains (Section 9.2), which is not a difference this evidence can resolve. The nested construction is retained because the variability it admits is variability the estimand contains and the fixed-model interval excludes by construction, not because it measures better here; and the price it charges is the subject of the next two paragraphs. The comparison a reader would want — against a *weighted* bootstrap, which removes that price at source — is below, and what it found is not what this construction's defence predicted.

Throughout, K denotes the **cardinality of the register field** — the number of distinct values f takes on a pair, 337 at the median pair and 5,245 at the largest (Table 3) — and n the number of training cases. The ratio K/n governs everything in this subsection and in Section 6.3.

Refitting inside the draw breaks the percentile interval. A bootstrap resample holds about $1 - e^{-1}$ of a high-cardinality categorical's distinct levels, so every refit sees a sparser register and returns a smaller increment; the bootstrap distribution is displaced and a percentile interval taken from it is a correct interval for the wrong quantity. On the sparse world of Section 9 it covers 37.2% against a nominal 95%, *worse* than the fixed-model interval refitting was meant to improve on. Every interval here is therefore the *basic* (pivotal) interval $[\,2\hat{V} - q_{1-\alpha/2},\ 2\hat{V} - q_{\alpha/2}\,]$ [10], and simultaneous bands are recentred by the same shift (Supplement S3.7).

The comparison Section 10 said was owed, and what it found. A previous version of this paper named the repair — drawing *weights* per moving block, so that every row has positive weight in every draw and every register level is present in every refit — and did not run it. It is run here: both schemes over one design, at one draw count, with one set of seeds, on 24,624 cells. **Every band this paper reports is the weighted one**; the multinomial arm appears only in this subsection, as the thing compared against.

The mechanism is removed completely. The share of the register's training levels a draw actually contains rises from 80.8% under multinomial resampling, 60.2% at worst, to 100% under weights, where it is one by construction. **And with it goes the starkest consequence, which this paper had never reported**: under the multinomial scheme a draw can drop an arm *entirely*, so the cell is absent rather than noisy, the median down its column is undefined, and the cell gets no band at all. Such a cell is not resolved — so it was counted as *unresolved*, beside cells that were banded and straddled zero, inside the denominator of every statement about how little the data resolve. On the previous surface there were 150 of them. Under weights there are 0. That pair changes the design as well as the scheme, so it credits the resampling with an improvement the balance may have supplied; the multinomial arm run on *this* design at the same draw count isolates the scheme, and it carries 270 such cells of 3,420. The clean comparison is the larger of the two.

The symptom, however, survives the removal of its supposed cause. A pivotal interval need not contain the estimate it is built from: pivoting moves the interval by twice the displacement, and where the displacement exceeds the interval's own half-width both endpoints land on the same side of $\hat{V}$. That is a property of a pivotal interval under bias rather than an error, but it is not what a reader expects of a band, so it is counted — and it falls from 965 cells (3.9%) to 322 (1.3%), which is better and is not zero. The displacement's own median barely moves, 0.0050 to 0.0041.

So the diagnosis this paper offered was wrong, and that is the finding. Level loss was named as the cause of the displacement. Removing level loss entirely leaves most of the displacement standing; therefore it was not the main cause, and this paper does not know what is. Section 10 carries that rather than a claim that the scheme is now unbiased. **The comparison is also not symmetric, and it leans the way that flatters the old scheme**: where a draw is missing, the band construction refuses the cell while the comparison drops the missing draws and computes from the rest, so every multinomial number above is conditioned on the draws in which the arm could be fitted — and those cells carry roughly twice the displacement of the others. Both multinomial numbers above are therefore measured on that scheme's successes, and both are still the larger of the pair.

Under the multinomial scheme it gets worse with the sample size, not better. That displacement is a bias — a multinomial resample holds about $1 - e^{-1}$ of the distinct levels however many rows it has — so at a fixed ratio it does not shrink with n while the half-width shrinks like $n^{-1/2}$. Section 9.3 locates the crossing: at a ratio of 0.125 the interval straddles its own estimate in every replicate up to 8,000 training rows and in only 1.6% of them by 50,000. This corpus reaches a ratio of 0.324 at its highest pair and 10 of 19 sit above a tenth, so it has pairs in that regime. **The argument does not carry to the bands reported here.** Weights remove the mechanism it rests on, and a displacement survives them anyway (above), so what remains under the reported scheme is not bounded by this reasoning and is not explained by it. Section 10 carries it as an open defect rather than a controlled one.

4.2. Three surfaces, and which one a claim may quantify over

A statement of the form *this surface is uniformly beneficial* is a statement about a denominator, and the denominator has to be the set the statement ranges over. Three sets are distinguished, named, and counted from the files rather than quoted (Table S14). The **declared surface** is every scientifically admissible specification: the product of the axis levels declared in Section 5. The **computational surface** is every specification actually evaluated; here it equals the declared surface — 29,040 scalar cells against 29,040 declared, with 0 duplicates and 0 missing — and the audit script asserts that identity rather than the manuscript asserting it. The **inference surface** is every specification carrying bootstrap draws; it is smaller for one reason, that a draw refits the whole pipeline and its cost is the declared surface's multiplied by the draw count.

The rule that selects it, in full. It is one design, and it is the same design on every pair: two learners, two splits, three register-quality conditions and three admissible rungs, crossed — 36 cells a pair, 3,420 scalar family members across the corpus — at 400 draws. Nothing in the rule reads a result, and nothing in it reads the log: the size classes that chose the previous surface's levels are gone, and the objection they carried with them.

It is a complete factorial, and that is checked rather than declared. On every one of the 19 pairs each combination of the four axes is present exactly once — no cell missing, no cell twice, and no pair differing in shape from any other — so a corpus median over these families is a median over one design, which the previous surface's was not. A balanced full factorial is also strictly stronger than the resolution-IV fraction that would suffice to make the decomposition estimable here, so the guarantee is the stronger one and not the cheaper one.

One objection this retires, and one it does not. The inference surface used to be *a corner of the declared surface rather than a design*, and the decomposition of Section 4.5 could not be computed on it. It can now, and is: one decomposition per pair per instrument, over 36 cells each. Supplement S14.2 reports what that shows, and it is not a confirmation — the axis carrying the largest first-order index agrees between the two surfaces on a minority of pairs. The two do not declare the same levels, so the disagreement compares two designs rather than showing either unrepresentative; what it does show is that a first-order index is a statement about the levels an analyst declared and not only about the pair.

The objection it does *not* retire is the one that matters most for reading a region label. The inference family is still a fraction of the admissible surface — a median 16.7% of a pair's admissible cells, and less than a twentieth on the pair with the richest grid — and **a label computed over fewer cells is systematically cleaner than one computed over all of them**. Section 4.6 says which label that most flatters, and Section 6.3 reports where it lands.

Every quantity that does *not* need a bootstrap — the decomposition, the sign-disagreement rate, the regret comparison — is computed on the full declared surface; only the region labels are computed on the inference surface.

4.3. Simultaneous bands over the family the claim ranges over

A decision curve read across 31 thresholds, a ladder read across rungs, or a surface read across every admissible cell, is a family of statements, and a pointwise interval does not control the probability that *some* member of the family is wrong. For a declared family $\mathcal{F}$ of cells with bootstrap draws $V_c^{(b)}$ and per-cell standard errors $\hat{\sigma}_c$, let

$$T^{(b)} \;=\; \max_{c\in\mathcal{F}} \frac{\left|V_c^{(b)} - \bar{V}_c\right|}{\hat{\sigma}_c}, \qquad q_{1-\alpha} = \text{the } (1-\alpha) \text{ quantile of } \{T^{(b)}\},$$

and report $\hat{V}_c \pm q_{1-\alpha}\hat{\sigma}_c$. This is the max-$t$ (Romano–Wolf) construction [31] and has family-wise coverage over $\mathcal{F}$ asymptotically, and over nothing larger.

The family must be the family. Within-cell bands, each covering five members, do not cover the several hundred cells a surface-level label ranges over. Four interval objects are declared and every table says which it prints: POINTWISE, WITHIN-INSTRUMENT, WHOLE-SURFACE — max-t over *every* admissible scalar cell of the pair, the only family a surface-level claim may use — and DECISION-CURVE, declared separately because net benefit and a scalar instrument are not the same scale. The whole-surface family has a median 180 members against five, and its median critical value is 3.36 against 2.37 and the pointwise 1.96. Family control is per log–target pair; every cross-pair statement here is descriptive. Supplement S3.3 gives these four *interval* objects — which are not the reporting objects of Table 1 — and the rule for dropping a replicate that leaves a cell without outcome variation.

Estimating the critical value without paying for it in refits. $q_{1-\alpha}$ is the upper quantile of a maximum over hundreds of correlated cells and the empirical quantile of B observed maxima estimates it badly, while raising B multiplies the cost of the whole design space. Standardising the draws and multiplying them by independent standard normals gives a statistic with their empirical covariance, generated 20,000 times for the cost of matrix arithmetic [5]; Supplement S3 gives the construction. The budget therefore fixes B and not the precision of q — order-statistic widths of 1.35 against 0.036 — and the residual Monte Carlo interval is carried through, so a cell counts as *resolved* only at its *upper* end. A max-t band also inherits the coverage of the interval it is built from, which Section 9.3 shows is governed by the ratio of register cardinality to training size. Supplement S9 derives a q calibrated at each pair's own ratio and Section 9.4 measures why it is not applied.

What these bands cover, measured against a known answer. **They do not attain their nominal level anywhere in this corpus's configuration.** Section 9.4 measures family-wise coverage over synthetic families matched to this corpus in size, draw count, cross-cell correlation and tails, whose truth is known by construction and whose bootstrap is *ideal*, so every coverage it reports is an upper bound on what the real procedure attains. On the family matched to the design this paper reports — this corpus's cell count, its draw count and its tails — the multiplier band's family-wise coverage is 91.6% against a nominal 95%, at a Monte Carlo standard error of 0.6%. Across the 7 surface families, which move the cell count and the draw count around it, it falls to 80.3%; on the decision-curve families it is a median 81.5% and a minimum 74.1%. It is 87.6% even under the Gaussian draws the construction is derived for. **The binding constraint is not the draw count**, which an asymptotic argument would suggest and this design's own measurement refutes: at the modal surface family coverage runs 81.2% at 33 draws and then *plateaus* — 91.2% at 150, 91.6% at 400 and 91.3% at 1,000. This corpus runs at 400 draws throughout, on the plateau and not below it, so the shortfall is not one a larger budget removes. Section 9.4 reports all five candidate critical values, the safeguard that buys almost nothing, and the shortfall in the units the calibration works in. **Every band, region label and ρ in this paper is therefore an anti-conservative statement**: they are reported as descriptive diagnostics, they are not coverage guarantees, and Section 10 states what follows for each conclusion that rests on one.

4.4. Planned contrasts, and a p-value the design can produce

Five contrasts on the primary log are *planned for this analysis round* — that the register adds to the intake baseline, that it still adds once the free field is admitted, that it adds at the later decision time, that the absorption D is positive, and that a half-empty register is worth less than a full one — each stated with its complete estimand in Table S15. **They are not confirmatory**: they were fixed before this analysis round, not before the data were seen, so the Holm-adjusted one-sided bootstrap p-values [16] they carry are reported for legibility and not as a confirmatory test. 5 of them reject at $\alpha = 0.05$ on the p-value — but **a contrast is called resolved only when the interval says so**, and on PC3 the one-sided p-value and the two-sided basic interval disagree, its interval containing zero. So the count that follows this paper's own rule is 4, and the p-value count is printed beside it rather than instead of it. The other detail is in Supplement S3.4: at 2,000 draws the smallest attainable Holm-adjusted value is 0.00250, because the plus-one estimator $\hat{p} = (k+1)/(B+1)$ is the only one a finite bootstrap can support. Every other comparison in this paper is *exploratory* and carries no multiplicity correction, because none would be meaningful over a surface enumerated in full.

4.5. The sensitivity decomposition

Because the surface is a complete factorial over its axes — the audit of Section 4.2 establishes that it is, cell for cell — the variance of V_s admits an exact and *orthogonal* functional-ANOVA decomposition [37, 32] under a product measure $\mathbb{P} = \bigotimes_i w_i$. Every subset u of axes has a component g_u, a Möbius sum over the conditional means computed exactly rather than estimated (Supplement S3); the components are orthogonal, so $\sum_{u \neq \emptyset} D_u = \mathrm{Var}(V)$ exactly, and with $S_u = D_u / \mathrm{Var}(V)$ the first-order and total indices are $S_i = S_{\{i\}}$ and $S_{T_i} = \sum_{u \ni i} S_u$. The closest precedent is Hutter et al. [17], who decompose an algorithm's performance over its *configuration* space by the same device; what differs here is that the axes are analysis decisions rather than hyperparameters, that the components are exact over a complete factorial rather than estimated from a surrogate over a sampled design, and that the decomposition is one of three reporting objects rather than the conclusion.

The split is an error stratum, and the primary decomposition treats it as one. Five of the split axis's six levels are expanding-origin folds on the same data. They are different *samples*, not different analyses, and a decomposition that pools them reports sampling variability as specification sensitivity. Because the decomposition is exact and orthogonal, the repair is a partition rather than a model: every component either involves the split or does not, so

$$1 \;=\; \underbrace{\textstyle\sum_{\emptyset\neq u\subseteq\mathcal{A}} S_u}_{\text{analyst choice}} + \underbrace{S_{\{\text{quality}\}}}_{\text{counterfactual}} + \underbrace{\textstyle\sum_{u\not\ni\text{split}} S_u - (\cdots)}_{\text{mixed}} + \underbrace{\textstyle\sum_{u\ni\text{split}} S_u}_{\text{resampling}}, \tag{3}$$

where $\mathcal{A} = \{\text{pipeline}, \text{rung}\}$ and the third term is what the split-free components carry beyond the first two, and the four terms are computed, not estimated (Table 5); the identity holds to 2.2×10^{-16}. The first three are what survives fold-averaging, and they are decomposed again *on* the fold-averaged surface, where each index is a share of the variance that fold-averaging leaves and the higher-order share is net of resampling. **Section 6.2 reports that decomposition as the primary one** and the pooled decomposition beside it, because the pooled one is what a reader of the earlier specification-curve literature will expect and the difference between them is itself a result.

The per-axis difference $S_{T_i} - S_i$ is the variance axis i is *involved* in beyond its main effect. These overlap — a two-way component appears in both its axes' total effects — so their largest member is not the share carried by interactions. That share is

$$S_{\text{int}} \;=\; 1 - \sum_i S_i, \tag{4}$$

which this paper computes together with every disjoint component, so the identity $\sum_u S_u = 1$ is asserted numerically — it holds on all 304 decompositions, which is the 19 pairs crossed with the 5 instruments under each of the three declared product measures, plus one per pair on the pooled cross-instrument scale — rather than assumed. The other quantity is retained as the *largest per-axis interaction involvement*; the two differ substantially, at 43.7% against 56.0% at the median pair.

4.6. Resolution regions and the robustness index

Definition 3 (Cell labels and surface regions)**.** *Let $\mathcal{F}$ be the declared inference family of Section 4.2 for a pair — the cells that carry bootstrap draws, a median 16.7% of that pair's admissible scalar cells. Under the simultaneous* *band of Section 4.3, a cell of $\mathcal{F}$ is* beneficial *if its lower band is above zero,* harmful *if its upper band is below zero, and* unresolved *otherwise. A surface is* uniformly beneficial *if every cell of $\mathcal{F}$ is beneficial;* conditionally beneficial *if some are and none is harmful;* uniformly harmful *and* conditionally harmful *in mirror image;* sign-changing *if there is at least one cell of each kind; and* unresolved *if none is resolved —* ***or if fewer than 5% of $\mathcal{F}$ is resolved at all****. The* robustness index *is*

$$\rho \;=\; \frac{\#\{\mathit{beneficial}\} - \#\{\mathit{harmful}\}}{|\mathcal{F}|} \;\in\; [-1, 1],$$

and every directional label is reported with the counts of beneficial, harmful and unresolved cells it rests on.

The minimum resolved share is not free. Without it a surface earns a direction from as few as two resolved cells of a hundred and eighty: *conditionally beneficial* then means "nothing resolved harmful", which on a family the data barely touch is a statement about the band's width and not about the register. 5 of the 12 pairs that carry a direction lose it under the condition (Table 6), and ρ is printed beside the triple everywhere because ρ alone cannot distinguish two resolved cells from ninety. The threshold is a declared convention; the labels at 0 and 7 per cent are in Table 6's source file, so a reader who prefers another can see what it costs.

The label is relative to $\mathcal{F}$ and not to the whole admissible set, and the direction of that is not neutral. A label computed on a sixth of a pair's cells is systematically cleaner than one computed on all of them, and *uniformly beneficial* is the label most helped by it: a claim about every cell is easier to sustain over fewer cells. The manuscript therefore never says a surface is uniformly beneficial without this restriction being in force, and Section 6.3 reports the one pair that reaches it together with the share of that pair's admissible cells the label was earned over — which is the smallest such share in the corpus.

The two harmful labels are not decoration, and what this corpus does with them is a case of the minimum share doing its work. Table S9, whose region column is the label *before* the minimum share is applied, carries 1 conditionally harmful and 0 uniformly harmful surfaces out of 19; that one conditionally harmful label rests on 4 resolved cells of 180, so Definition 3 withdraws it, and under the definition this paper reports the count of either kind is 0. The label is available and the data do not earn it. $\rho = 1$ is the only state in which one positive number is a safe summary of a surface, and the empirical question this paper answers is how often it obtains. The region is defined over the *scalar* family; folding the decision-curve cells into it would let a single extreme threshold decide the label of a whole surface, so the decision curve gets its own region (Section 8). The intercept-only rung is excluded from every admissible set, because a reduction measured against a model with no features is inflated by construction; the master surface file *does* carry that rung, which is why the two counts in Table S14 differ.

4.7. Specification regret, as a robustness check

This subsection defines a check and not a reporting object. What *specification regret* reports is a null — the one-number rule's median excess regret is zero in every instrument, the out-of-sample ordering reverses under rolling folds (Section 6.6), and on calibrated net benefit what separates the rules is the conservative one rather than the one-number one (Section 8.4) — and a null is a check on the paper's other two arguments rather than a result of its own. It is reported in full, and it is not a contribution.

The construction asks what choosing one specification costs against the best one, on a scale a decision can use. Three designs are declared and all three reported: **A**, metric-specific, so that no average crosses an instrument; **B**, on calibrated net benefit, which is a common operational utility; and **C**, partially identified over a declared set of monotone utility maps, which reports an interval rather than assuming one map. The identity map's maximum excess is exactly zero. Supplement S3.1 gives the three constructions and the four decision rules compared under them.

A reporting standard follows from the apparatus, and is short enough to state. Report the design space and the measure over it; the denominator the claim quantifies over, named as one of the three; the absolute increment at every rung rather than a ratio; the decomposition with $1-\sum_i S_i$ reported as such; the region and ρ under the band's own family, with the counts they rest on; and the reduction R last if at all, and only where its denominator is resolvably positive. Supplement S16 states the standard in full, together with the errors this study made before it got them right, a six-step recipe, the `fieldvalue` package [11] that implements the objects and refuses by construction to emit a single number, and a worked example on a dataset that is not an event log at all.

5. The registered benchmark design

5.1. What was fixed, and when

The rules that turn an arbitrary event log into an instance of this problem were committed before any result on a new log existed, in `PROTOCOL.md` for the corpus and `AUDIT-PROTOCOL-2.md` for the practice pilot. The version history shows the registering commit adding one file and no result file; `git log --stat` is the check, and `REPRODUCE.md` names the commit. Six amendments were declared afterwards, each with its date, its cause and the result it changed, and all six are in Supplement S1 rather than folded silently into the rules.

This is not a timestamped third-party registration, and the difference matters. A registration on a public registry is attested by a party with no stake in the result; what is offered here is a file in a repository whose history the authors control, so a reader who does not trust that history has nothing else to check. Any claim in this paper that a rule was fixed in advance is therefore a claim about the repository and no stronger than the repository. We say so plainly instead of letting the word *registered* carry a weight it has not earned: what the design buys is that the rules are legible, dated and amendable only on the record, not that an outside party watched them being written.

5.2. Roles

A log is admitted if the registered rules can find three things in it without reading an outcome.

The register f. A high-cardinality attribute that is not resource-like, has at least 50 distinct values, is reused across at least two cases on average, and — by amendment 1 — whose values recur across the temporal split, so that a document number is not mistaken for a maintained entity.

The free field g. A resource-like attribute, present on at least 50% of cases, whose value varies within a case, so that it can carry information about routing. **Only the *first* value it takes on a case is used as a feature**, while the handover target is defined from the whole sequence of values — so the feature is what is known when the case opens and the target is what happens afterwards, and the field cannot inform the model about its own future. The distinction is not cosmetic: the

log	target	ITSM	register field f	free field g	axes: pipe/split/qual/rung	cases	levels of f
BPIC13_closed	handover	yes	product	org:resource	2/6/6/3	1,487	337
BPIC13_incidents	duration	yes	product	org:resource	2/6/6/3	7,554	704
BPIC13_incidents	handover	yes	product	org:resource	2/6/6/3	7,554	704
BPIC14	duration	yes	CI Name (aff)	assignment group	4/6/10/4	46,606	3,019
BPIC14	handover	yes	CI Name (aff)	assignment group	4/6/10/4	46,606	3,019
BPIC15_1	duration	–	case:parts	org:resource	2/6/6/3	1,199	89
BPIC15_1	handover	–	case:parts	org:resource	2/6/6/3	1,199	89
BPIC15_3	duration	–	case:parts	org:resource	2/6/6/3	1,409	94
BPIC15_3	handover	–	case:parts	org:resource	2/6/6/3	1,409	94
BPIC15_4	duration	–	case:parts	org:resource	2/6/6/3	1,053	53
BPIC15_4	handover	–	case:parts	org:resource	2/6/6/3	1,053	53
BPIC15_5	duration	–	case:parts	org:resource	2/6/6/3	1,156	107
BPIC17	duration	–	case: RequestedAmount	org:resource	2/6/6/3	31,509	701
BPIC19	duration	–	case:Vendor	org:resource	3/6/6/3	251,734	1,975
Helpdesk	duration	yes	customer	Resource	2/6/6/3	4,580	394
RoadFines	duration	–	article	org:resource	3/6/6/3	150,370	66
Sepsis	duration	–	Diagnose	org:group	2/6/6/3	1,050	147
UCI498	duration	yes	caller_id	assigned_to	2/6/6/3	24,918	5,245
UCI498	handover	yes	caller_id	assigned_to	2/6/6/3	24,918	5,245

Table 3: ROLES AND LEVELS, PER PAIR. Which recorded attribute plays the register f and which the free field g on each admitted pair, whether the pair is IT service management, and how many levels each varied axis carries: pipeline, split, register-quality condition, baseline rung. Every pair also carries the 5 scalar instruments, so the per-pair factorial is 5 axes wide and not the nine of Definition 1; Section 5 says which axes are fixed on which pairs and why. 'cases' is the admitted case count and 'levels of f' the register's cardinality. This table was Supplement Table S11 through round twenty-five; the master table cannot be read without it.

same attribute supplies both, and using the case-level set would put the answer in the features.

The intake block B_0. Every remaining attribute of cardinality at most 200 that is neither an outcome, nor a timestamp, nor a relabelling of the activity, nor a deterministic coarsening of f.

Coarsenings of f — its type, its subtype, the service it belongs to — are given their own role rather than being swept into B_0, because which *layer* of a register pays is a separate question, answered in Section 7.4.

Table 3 is what the rules found, and it is also the design space as it was actually varied. It names, per pair, the attribute that plays each role and the number of levels each axis carries. Two things are visible there that Definition 1 and Table 2 do not say. The *target* is not an axis of any surface:

the surface is computed per log–target pair, so each of the two targets a log carries has a surface of its own, and the cohort $\times$ target design exists only in Section 7.1. And the per-pair factorial is 5 axes wide — pipeline, split, register quality, baseline rung, instrument — not the nine of Definition 1: the decision time varies only on the case study's log, the operating point only in the decision-curve family, and the cohort only in Section 7.1. Definition 1 declares what the estimand's arguments *are*; Table 3 declares which of them this study *varied*, and Section 10 carries the difference as a limitation rather than leaving a reader to find it.

5.3. Targets, splits and exclusions

Two targets are defined for every log — *handover*, whether more than one resource group touches the case, and *duration*, whether it runs longer than the training half's median. Cases are ordered by first timestamp and split 70/30, with five expanding-window rolling-origin folds beside it, and fields determined only at closure are excluded by name [20, 19, 49]. A pair is excluded only for a registered reason, and **every admitted pair is analysed and reported** — running the surface only where the reduction resolves would condition the analysis on the outcome. Supplement S1.3 gives the rules, the exclusion codes and the admission margin at the upper edge of the prevalence window.

Two exclusions a reader of Table 3 will ask about. BPIC15_2 is excluded under SMALL_N at 832 cases while its four siblings are admitted — the rule is a floor on the case count and nothing about the log's content — and the further BPIC15 targets, the BPIC20 logs and BPIC12 are excluded under NO_HEADROOM or NO_G, which are prevalence and role failures. **And the registered handover target on the case study's log sits at the upper edge of the admission window**: test prevalence 0.927 against a window that closes at 0.95, which excludes other logs at 0.959. That pair carries 4,800 of the corpus's 29,040 admissible cells — the log carries 9,600 across its two targets — so a corpus statistic pooled over cells is partly a statement about it; Section S14.17 reports every headline on the ITSM family separately for this reason, and Section 10 asks whether the window should be tightened.

5.4. What the corpus does and does not have in common

The corpus is 13 public logs spanning 6 domains, and it includes the further ITSM logs the literature uses [39, 1]. It is heterogeneous and the paper does not claim otherwise. The registers are a configuration item, a caller, a product, a vendor, a diagnosis and a legal article; the free fields are an assignment group, an assignee, a resource and a role; the targets mean different things in a bank's incident queue, a hospital's admissions and a municipality's permits.

Table S13 makes that explicit rather than leaving it to a sentence. For every pair it names the domain, the field playing the register role, why that field behaves as a maintained and reused entity, the free field, the interpretation of the target, and the specific threat to construct validity that the pair carries. A diagnosis code is admitted as a register by a cardinality-and-reuse rule, and it is not a configuration management database; a legal article is reused across cases but nobody maintains a discovery process for it. Those are threats, they are named per pair, and the paper's conclusions are stated at a level that survives them.

This is therefore a *benchmark of specification sensitivity across heterogeneous prediction problems*, not a replication of one finding on several organisations, and two disciplines follow. No domain-specific effect magnitude is averaged across domains as though it estimated one universal register value: the corpus-level statistics are all about *sensitivity*, which is a property each pair has separately and which can be summarised across pairs without assuming they measure one thing. And the 8 ITSM-like pairs, the closest family to the case study, are analysed as a group as well, with *every* headline corpus statistic reported on both populations in Section S14.17.

6. Multi-log results

Every point estimate in this section comes from one master surface — one file, named in the code-availability statement — computed by one pass of one pipeline over 19 log–target pairs, 4 learners, 6 splits, 10 register-quality conditions, the nested baseline ladder and 36 instruments including the 31-point decision-curve grid. Its 273,024 rows *include* the intercept-only rung, so that the surface is complete; no admissible set contains it, and the 29,040 admissible scalar cells every headline is computed on exclude it. Every interval is the basic construction of Section 4.1 and every band the whole-surface simultaneous construction of Section 4.3, and each table names which it prints. The denominator audit is in Table S14: declared equals computed on all 19 pairs, with 0 duplicates and 0 missing, and the inference surface (3,420 scalar members) is a subset of it cell for cell, which `verify_numbers` asserts.

log	target	V at reference	pointwise	simultaneous	region	resolved share	rho	misreport rate
BPIC13_closed	handover	-0.009	[-0.042, +0.018]	[-0.062, +0.040]	unresolved	0.0	0.000	0.284
BPIC13_incidents	duration	0.028	[+0.009, +0.055]	[-0.007, +0.072]	conditionally beneficial	10.6	0.106	0.393
BPIC13_incidents	handover	0.037	[+0.028, +0.049]	[+0.021, +0.057]	conditionally beneficial	47.8	0.478	0.111
BPIC14	duration	0.096	[+0.089, +0.108]	[+0.082, +0.117]	uniformly beneficial	100.0	1.000	0.106
BPIC14	handover	0.167	[+0.146, +0.207]	[+0.117, +0.225]	conditionally beneficial	99.4	0.994	0.244
BPIC15_1	duration	0.001	[-0.036, +0.036]	[-0.064, +0.059]	unresolved (below minimum share)	2.8	0.028	0.423
BPIC15_1	handover	0.007	[-0.007, +0.020]	[-0.015, +0.031]	unresolved (below minimum share)	1.1	0.011	0.242
BPIC15_3	duration	0.035	[-0.006, +0.068]	[-0.030, +0.101]	conditionally beneficial	15.0	0.150	0.191
BPIC15_3	handover	-0.003	[-0.050, +0.052]	[-0.085, +0.081]	unresolved (below minimum share)	4.4	-0.022	0.616
BPIC15_4	duration	0.035	[+0.009, +0.079]	[-0.014, +0.105]	unresolved	0.0	0.000	0.623
BPIC15_4	handover	0.001	[-0.021, +0.042]	[-0.050, +0.063]	unresolved (below minimum share)	3.9	-0.017	0.618
BPIC15_5	duration	0.032	[-0.020, +0.103]	[-0.074, +0.138]	unresolved (below minimum share)	1.7	0.017	0.328
BPIC17	duration	0.002	[-0.004, +0.013]	[-0.010, +0.018]	sign-changing	12.2	0.056	0.353
BPIC19	duration	0.024	[+0.011, +0.040]	[-0.001, +0.052]	sign-changing	63.9	-0.406	0.360
Helpdesk	duration	-0.020	[-0.045, -0.002]	[-0.062, +0.015]	unresolved (below minimum share)	2.2	-0.022	0.257
RoadFines	duration	0.031	[+0.024, +0.038]	[+0.019, +0.043]	sign-changing	51.1	0.400	0.173
Sepsis	duration	0.153	[+0.102, +0.199]	[+0.073, +0.235]	conditionally beneficial	62.2	0.622	0.056
UCI498	duration	-0.007	[-0.014, +0.005]	[-0.019, +0.012]	conditionally beneficial	12.2	0.122	0.493
UCI498	handover	-0.003	[-0.007, -0.001]	[-0.009, +0.001]	unresolved (below minimum share)	3.9	0.039	0.481

Table 4: The master table. One row per admitted log–target pair. 'pointwise' is the 95% interval at the reference cell, basic (pivotal) construction; 'simultaneous' is that same cell's WHOLE-SURFACE simultaneous band, at the conservative end of the critical value's Monte Carlo interval. They are different objects, and the region label uses only the second. Then the resolution region and the robustness index ρ. NO COVERAGE CALIBRATION IS APPLIED TO EITHER: the (n, K) factor was fitted to restore POINTWISE coverage on a plane built around a different inference surface, and Section 9.4 measures the family-wise widening these families need as LARGER than it supplies, so these are the nominal labels and they are anti-conservative by an amount the paper reports. Table S9 prints what applying that factor would have given, beside these. THE REGION IS THE LABEL Definition 3 YIELDS, minimum resolved share included: 'resolved share' is the percentage of the inference family the band resolves, and a direction is withheld below 5% of it, which withdraws the direction on 5 of the 12 pairs that carry one. Table 6 MARKS 7 rows, which is the larger number: a pair the band already left unresolved is not marked, and 2 of the marked rows are sign-changing, which carry no direction to withdraw. Last, the share of admissible specifications whose sign disagrees with a conventional one-number report, under the equal-level measure, which is the 'all-cells rate' column of Table S10. The reference cell's sign and the region need not agree, and on several pairs they do not: a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative.

6.1. The master table

Table 4 is the frozen master table: one row per log–target pair, the increment at the reference specification with its simultaneous band, the region label, the robustness index and the sign-disagreement rate of a conventional one-number report. The reference specification is declared and is the same everywhere — one-hot logistic regression, the single temporal holdout, the clean register, the intake block plus the free field as the baseline, ROC AUC — and its region and ρ columns are the ones Section 6.3 reports while its sign-disagreement column is the all-cells rate of Table S10 under the equal-

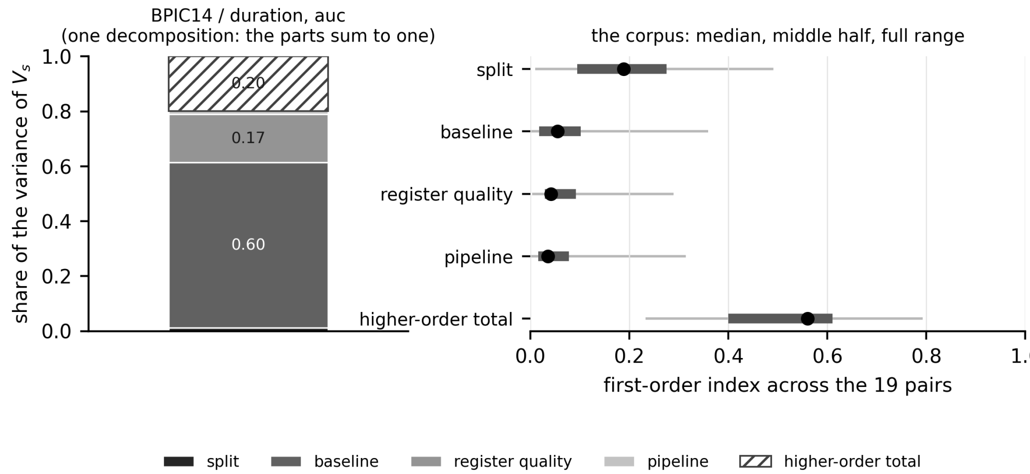

Figure 3: The decomposition, within instrument, under the equal-level measure. *Left*: one pair at one instrument, so the components are a decomposition and the parts sum to one exactly — the largest pair in the corpus, at the reference instrument, which is the cell every other table reports at. The hatched segment is the total higher-order share $1-\sum_i S_i$ of (4). *Right*: each axis's first-order index across the corpus, as the median over pairs with the range the middle half occupies and the full range. Nothing on the right is stacked, because a median over pairs and a median over instruments do not add and a stacked bar of them would invite the addition. The bootstrapped indices and their intervals are in Table S8.

level measure. **The reference cell's sign and the region label need not agree, and on several pairs they do not**: a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative, because the label ranges over the whole inference family and the reference cell is one member of it. Table 6 prints the counts each label rests on, and Table 3 which attribute plays the register on each row.

6.2. What the choices are worth

Table S8 and Figure 3 give the decomposition, computed within each instrument in that instrument's own units, under the equal-level measure. The table's first four columns are, per pair, the median over the five instruments of each axis's first-order index; its last column is the largest per-axis interaction *involvement*. Three aggregations of this table are quoted below and they are not interchangeable, so each is named where it appears: the median over pairs of a column of this table; the 28.7% figure, which is the median over pairs of the median over instruments of the *per-instrument* largest first-order

log	target	analyst choice	counterfactual	mixed	resampling
BPIC13_closed	handover	31.5	2.9	5.8	59.8
BPIC13_incidents	duration	13.3	2.9	3.0	79.6
BPIC13_incidents	handover	19.2	9.9	4.5	42.3
BPIC14	duration	71.9	15.1	7.9	5.1
BPIC14	handover	73.9	12.7	10.3	3.6
BPIC15_1	duration	9.4	3.9	4.7	80.5
BPIC15_1	handover	13.4	3.6	3.6	62.6
BPIC15_3	duration	11.7	8.1	1.5	77.6
BPIC15_3	handover	22.3	0.4	2.5	76.1
BPIC15_4	duration	11.3	2.5	5.6	81.6
BPIC15_4	handover	12.4	3.1	5.7	71.7
BPIC15_5	duration	17.4	4.2	3.9	69.9
BPIC17	duration	4.2	6.2	5.2	84.1
BPIC19	duration	3.8	1.1	2.8	90.9
Helpdesk	duration	31.7	8.6	1.7	53.6
RoadFines	duration	7.0	28.9	2.1	61.7
Sepsis	duration	13.2	14.5	5.0	71.9
UCI498	duration	11.0	3.0	4.3	81.1
UCI498	handover	41.0	7.1	9.3	42.4

Table 5: THE DECOMPOSITION WITH THE SPLIT AS AN ERROR STRATUM. Per cent of the pooled variance of V_s, per pair, median over the 5 instruments, under the equal-level measure. The four columns are an exact partition of the orthogonal decomposition's components, and on *each* of the 304 decompositions they sum to 100 to within 2.2×10^{-16}. **The printed rows are medians over instruments and therefore need not sum to 100**, for the same reason Figure 3's right panel is not stacked: a median is not linear. 'analyst choice' collects every component built only from the pipeline and the baseline rung, 'counterfactual' the register-quality main effect, 'mixed' the components that join the two, and 'resampling' every component involving the split — five of whose six levels are expanding-origin folds on the same data. Dropping the sixth, the single temporal holdout, so that the axis is folds and nothing else, moves the last column's corpus median from 71.7% to 71.0%.

index, and which is therefore not obtainable by taking a maximum across the four printed columns; and the corpus medians of the higher-order share. The per-instrument quantities the second is built from are in Table S26.

Most of what the pooled surface varies over is not specification at all. Table 5 partitions the decomposition by (3), exactly and on every decomposition separately. Taking each column's median across the corpus — *four medians, which do not add, and are not four parts of one pair* — the components involving the split carry 71.7% of the variance, against 13.3% for the choices an analyst makes, 4.2% for the register-quality counterfactual and 4.5% for the components that join them. Resampling is the largest of the four on 17 of the 19 pairs. Dropping the one split level that is a genuine analyst choice — the single temporal holdout, leaving five folds of one rule — moves that share only from 71.7% to 71.0%, so the stratum is not an artefact of mixing a rule with its folds. **The two exceptions are the case study's own pairs**, where the analyst axes carry the richest grids in the corpus — 4 pipelines and 10 register-quality conditions, against at most 3 and 6 on every other pair (Table 3) — and the ordering reverses.

Net of resampling, no single axis dominates — and this is a weaker statement than the pooled surface makes it look. On the fold-averaged surface, which carries 28.3% of the pooled variance, the median first-order index is 21.9% for the baseline rung, 19.2% for the register-quality condition and 15.2% for the pipeline, and at the median pair the largest first-order index *within an instrument* is 38.8% — a median over pairs of a median over instruments of the per-instrument largest, and so not the largest of the three medians just quoted. **The total higher-order share is 26.3% once resampling is removed, against 56.0% on the pooled surface, and it no longer exceeds the largest first-order index**: pooled, the interactions carry more than any main effect on 13 of 19 pairs; net of resampling, on 7. *Most of the variance belongs to no single axis* is a property of a surface that counts five folds as five specifications, and it does not survive treating them as what they are. What does survive is the weaker and still sufficient claim: which axis leads is a property of the pair — the rung on 8, the register-quality condition on 5 and the pipeline on 6 — so an analyst cannot learn from somebody else's paper which of their own choices their answer turns on.

The pooled decomposition, for comparison. Across the corpus the median pooled first-order index is 18.9% for the split, 5.5% for the baseline rung, 4.2% for the register-quality condition and 3.6% for the pipeline; at the median pair the largest first-order index *within an instrument* — the second of the three aggregations named above, and not the largest of those four medians — is 28.7%, the total higher-order share is 56.0% (95% interval 40.9% to 60.8% over pairs, 23.3% to 79.3%), and the largest per-axis *involvement* $\max_i(S_{T_i} - S_i)$, a different quantity, is 43.7%. It is reported because it is what the specification-curve literature computes and because the difference between the two is this subsection's result, not because it answers the question as asked.

The measure is not neutral, and one conclusion does not survive it. Across the three product measures the higher-order share moves by 24.3%, and under the measure concentrated on the reference cell it is 31.7% — smaller than the largest first-order index under the same measure, 39.2%, on 12 of 19 pairs. *The interactions carry more than any main effect* is therefore a claim about a measure and not only about a surface. Supplement S14.11 carries the target axis, which axis leads on which pair, and both invariance tests.

The baseline axis moves the answer by more than the answer, on the pairs where an analyst has more than one baseline to choose between. Holding the pipeline, the split, the register and the metric at the reference and varying only the baseline rung, the increment moves by 0.073 AUC at the median pair and 0.252 at the widest — larger than the increment itself at the reference cell on 15 of the 19 pairs. **At the median pair that range is carried by the half-of-intake rung — at the widest pair it is not — and a reader is owed the spread without it.** Half of an intake block, taken in a deterministic order by cardinality, is admissible but impoverished in the same way the intercept-only rung this paper already excludes is impoverished. Excluding it too, the spread is 0.0055 AUC at the median pair and 0.2495 at the widest, and exceeds the reference increment on 7 pairs. **How much of the range that rung carries is itself a property of the pair**: dropping it cuts the spread by a median 88.1% per pair, by 1.0% on the pair holding the corpus's widest spread, and not at all on 2 pairs, where the rung's own increment falls inside the range the remaining rungs already span. The fall is not a correction of an error but a change of denominator with a plain explanation: on 17 of the 19 pairs only two rungs remain, so the spread over them *is* the absorption D of the free field, which Section 7.5 reports as an object in its own right with its own interval. Only the case study's log carries a third (Table 3), and its two pairs are where the corpus's widest realistic-rung spread sits, reaching 0.2495. The index and the spread answer different questions — what share of the *variance* an axis explains while every other axis is also varying, against how far that one axis moves the answer on its own — and both are reported because either alone misleads.

The pipeline axis is two axes, and it does not run at full width everywhere. Its levels are a model family together with an encoding and, in one case, a calibration, and on 15 of the 19 pairs only two of them run. **The admission rule is a function of the log's row count and nothing else**, declared before any fit. On the declared surface the case study's log carries all 4 levels, the two largest logs carry 3 and every other pair 2, which Table 3 gives per pair. **Inside the inference family that rule no longer applies**: every pair carries the same two levels, differing in model *family*, with the boosting arm retained everywhere (Section 4.2). The fused index below is therefore a statement about the same axis on every pair rather than about a different one on each. Crossed on the case study's log — both families against all three encodings, 6 complete factorial pipelines, run on each of the log's two targets, which index surfaces rather than enlarging one (Section 5) — **the encoding is the larger of the two**, 10.1% of the variance against the family's 5.2%, each the median over those two targets and the 5 instruments pooled, with an interaction, 9.7%, that lies between the two main effects and is the largest two-way term: the fused index is not the sum of two indices. Supplement S14 carries the crossed decomposition, and Table S7 the per-target medians, which are a different aggregation and need not agree with these.

Run on the 8 ITSM pairs, that finding travels but does not hold everywhere. Taking the median over the 5 instruments within each pair — a third aggregation, named here because the two above are not it — the encoding is the larger first-order axis on 6 of the 8, and on the other 2 **the family is, and not narrowly**: on the more extreme of them the encoding carries 0.04 times what the family does. So the case study's ordering is the common one and not the rule. **That is this section's own thesis arriving where it was not sent**: the crossing was run to separate two axes a fused index confounds, and what it finds is again that which axis leads is a property of the pair. The remaining pairs are not crossed, which Section 10 records.

6.3. Resolution regions, and the coverage they have

Table S9 and Figure S4 carry the region labels and the robustness index for every pair, computed on the inference surface under the whole-surface simultaneous band at the conservative end of the critical value's Monte Carlo interval. That family has a median 180 members per pair and a median critical value of 3.36, against 2.37 for a within-instrument family and 1.96 pointwise; under the narrower family the same draws give a different region label on 8 of the 19 pairs, and the whole-surface family resolves 890 cells against the narrow family's 1,135.

The nominal critical value is not enough, and this paper does not have a correction that closes the gap. Section 9.3 shows that pointwise coverage is governed by K/n and that this corpus lives where it is below nominal, the median pair sitting at 0.106, so a region label computed from a nominal band overstates how much of a surface is resolved. The (n, K) plane is dense enough to calibrate against, and Supplement S9 derives an inflation factor $c(K/n)$ from it — a monotone regression through the measured factors, whose log-linear summary has slope 0.102 (standard error 0.005), and which on this corpus would supply widenings of 1.18 to 1.77.

That factor is not applied to the labels this paper reports, and the reason is a measurement rather than a preference. It was fitted to restore *pointwise* coverage, on a plane built around the previous inference surface. Section 9.4 measures the widening the *family-wise* band actually needs on families matched to the surface reported here, and it is 1.36 to 1.88 — *larger* than the factor would supply, because these families carry heavier tails (a median per-cell excess kurtosis of 0.52 and 7.9 at the ninetieth percentile). Applying a widening known to be under-sized, and presenting the result as the operative band, would buy the appearance of a correction and not the correction. **The labels below are therefore computed at the nominal critical value, and they are anti-conservative by an amount this paper has measured and cannot remove**: 890 cells resolve corpus-wide, and a band wide enough to cover would resolve fewer.

Supplement S9 carries the factor's reach in n, the evidence that K/n alone is not sufficient, and the 4 of 33 cells on which the factor does not exist at all because past a crossing in n the basic interval stops containing the point estimate a widening would be about; Section 10 carries what that leaves the corpus's largest pairs.

Of the 12 pairs that resolve anything, 10 resolve cells of one sign only — 8 beneficial and 1 harmful — and 2 resolve cells of both. That is the informative reading of the region column, and it is the one to carry away: on 10 of the 12 pairs where the data determine any sign at all, every sign they determine points the same way; on the remaining 2 they do not.

One surface of the 19 is uniformly beneficial, over a family that is 3.75% of that pair's admissible cells. Section 4.6 poses the question this answers: $\rho = 1$ is the only state in which one positive number is a safe summary of a surface, and how often it obtains is an empirical matter. It obtains 1 time here, on the pair carrying the richest declared grid in the

log	target	family size	beneficial/harmful/unresolved	resolved share	region	label under minimum share
BPIC13_closed	handover	180	0/0/180	0.0	unresolved	–
BPIC13_incidents	duration	180	19/0/161	10.6	conditionally beneficial	–
BPIC13_incidents	handover	180	86/0/94	47.8	conditionally beneficial	–
BPIC14	duration	180	180/0/0	100.0	uniformly beneficial	–
BPIC14	handover	180	179/0/1	99.4	conditionally beneficial	–
BPIC15_1	duration	180	5/0/175	2.8	conditionally beneficial	unresolved (below minimum share)
BPIC15_1	handover	180	2/0/178	1.1	conditionally beneficial	unresolved (below minimum share)
BPIC15_3	duration	180	27/0/153	15.0	conditionally beneficial	–
BPIC15_3	handover	180	2/6/172	4.4	sign-changing	unresolved (below minimum share)
BPIC15_4	duration	180	0/0/180	0.0	unresolved	–
BPIC15_4	handover	180	2/5/173	3.9	sign-changing	unresolved (below minimum share)
BPIC15_5	duration	180	3/0/177	1.7	conditionally beneficial	unresolved (below minimum share)
BPIC17	duration	180	16/6/158	12.2	sign-changing	–
BPIC19	duration	180	21/94/65	63.9	sign-changing	–
Helpdesk	duration	180	0/4/176	2.2	conditionally harmful	unresolved (below minimum share)
RoadFines	duration	180	82/10/88	51.1	sign-changing	–
Sepsis	duration	180	112/0/68	62.2	conditionally beneficial	–
UCI498	duration	180	22/0/158	12.2	conditionally beneficial	–
UCI498	handover	180	7/0/173	3.9	conditionally beneficial	unresolved (below minimum share)

Table 6: WHAT A REGION LABEL RESTS ON. $|\mathcal{F}|$ is the inference family's size; the triple counts its cells the whole-surface band calls beneficial, harmful and unresolved. A directional label follows from as few as two resolved cells, and the last column applies the minimum resolved share of Definition 3 — 5% of the family — which withdraws the direction on 5 of the 12 pairs that carry one. A dash means the label is unchanged.

corpus — and **the denominator belongs in the same breath as the label**. The state is reached over an inference family of 180 cells against that pair's 4,800 admissible ones — the smallest share in the corpus, which 2 pairs attain jointly rather than this one alone — and the label requires every cell of *that* set to resolve beneficial (Definition 3) — a claim about every cell being correspondingly easier to sustain over fewer of them. **So the one case in which a single number is safe is a case where the claim ranges over 3.75% of the specifications an analyst could have chosen**, which is a statement about a denominator rather than a counterexample to one. It is also not an artefact of the band's width: the label survives every widening the measured shortfall of Section 9.4 could ask for, and Supplement S14.18 gives the whole reckoning. The counts are reported at four calibration settings and the ordering is stable across all of them (Supplement S9.7), so no conclusion here rests on the calibration being exactly right.

6.4. What one number costs

A conventional report — one pipeline, the clean register, the single hold-out, ROC AUC — disagrees in *sign* with a median 32.8% of the other admissible specifications under the equal-level measure. **Three things are wrong with reading that as *what a one-number report exposes a reader to***, and this section corrects each of them, then all three at once: the axes are not all the same kind of thing, most cells are not cells whose sign the data determine, and a disagreement counted at any magnitude is not a disagreement worth acting on. Correcting them one at a time is not enough — each correction moves the denominator the next one is computed on — so the rate that survives is the one taken under all three together.

The axes are not all the same kind of thing. The design space mixes choices an analyst makes with things that are not choices at all, and Table S20 partitions the decomposition accordingly: *analyst latitude* is the pipeline and the baseline rung; *resampling* is the split, five of whose six levels are expanding-origin folds on the same data; *counterfactual* is the register-quality condition, a different state of the world rather than a different way of analysing this one. On the primary scale, analyst latitude carries a median 11.1% of the variance, resampling 18.9% and the counterfactual 4.2%, and resampling is the larger of the first two on 10 of the 19 pairs.

Whether the split moves the increment more than the analytic axes depends on which scale the partition is taken under, and we report both rather than the one that reads better. The primary scale decomposes *inside* each instrument and takes the median across the five, so the instrument is a stratum and its own contribution is in none of the three columns. Making it a sixth axis needs the headroom rescaling of Section 4.5, because raw AUC and raw average precision do not share units — and on that scale the ordering reverses: analyst latitude carries 19.2%, resampling 9.4% and the counterfactual 2.7%, with resampling the larger on 7 pairs. The instrument is an analyst's choice, so the second number is the one that answers the question as asked; the first is the one taken on the units the rest of Section 6 uses. Neither is a corpus fact independent of the scale, which is the paper's own thesis applied to the paper. Restricted to the analyst-latitude cells — the pipeline by the admissible rungs by the 5 instruments, with the register clean and the split at the single holdout, a median 30 cells per pair and 700 over the corpus, because the pipeline and the rung do not carry the same number of levels on every pair (Table 3) — the median per-pair disagreement rate is 20.0% against the all-cells 32.8%, both medians over pairs: the first is what a one-number report exposes a reader to, the second what a claim ranging also over degraded registers and resampled splits is exposed to.

The reference specification is itself a choice, and moving it moves the headline by less than the corrections above. The rate counts cells whose sign disagrees with the reference cell, so the reference supplies the sign and nothing else. Varying it one axis at a time over every level that axis

log	target	V at reference	above MPID	ref. resolved	all cells	above MPID rate	resolved	resolved and above MPID	mean deviation, AUC
BPIC13_closed	handover	-0.0091	–	–	39.4	–	–	–	–
BPIC13_incidents	duration	0.0278	yes	–	21.8	16.2	0.0	0.0	0.0601
BPIC13_incidents	handover	0.0373	yes	yes	8.3	5.5	0.0	0.0	0.0510
BPIC14	duration	0.0959	yes	yes	8.8	0.9	0.0	0.0	0.0204
BPIC14	handover	0.1673	yes	yes	23.3	21.4	0.0	0.0	0.0623
BPIC15_1	duration	0.0011	–	–	38.4	–	–	–	–
BPIC15_1	handover	0.0071	–	–	22.7	–	–	–	–
BPIC15_3	duration	0.0354	yes	–	7.4	4.8	0.0	0.0	0.1047
BPIC15_3	handover	-0.0030	–	–	74.1	–	–	–	–
BPIC15_4	duration	0.0348	yes	–	56.5	56.3	–	–	–
BPIC15_4	handover	0.0015	–	–	70.8	–	–	–	–
BPIC15_5	duration	0.0320	yes	–	21.8	17.6	–	–	–
BPIC17	duration	0.0021	–	–	23.1	–	0.0	–	0.0221
BPIC19	duration	0.0244	yes	–	17.6	16.4	0.0	0.0	0.0528
Helpdesk	duration	-0.0195	yes	–	37.5	32.4	–	–	–
RoadFines	duration	0.0306	yes	yes	4.0	2.9	0.0	0.0	0.0129
Sepsis	duration	0.1527	yes	yes	0.0	0.0	0.0	0.0	0.0335
UCI498	duration	-0.0069	–	–	61.1	–	–	–	–
UCI498	handover	-0.0033	–	–	54.2	–	–	–	–

Table 7: SIGN DISAGREEMENT ONCE MAGNITUDE IS REQUIRED. On the AUC sub-surface, because the minimal practically important difference is declared in AUC and Remark 1 forbids carrying it to another instrument. The four rate columns are per cent of cells whose sign disagrees with the reference cell: over every admissible AUC cell; over those where the reference increment and the cell's own increment both exceed the MPID of 0.01 AUC; over those the whole-surface band resolves; and over cells that satisfy both restrictions. A dash is not a zero — it is a restriction that leaves no cell to compute a rate on. The last column is a magnitude and not a rate: the mean distance from the reference increment over the cells the band resolves, in AUC.

carries — 26 variants in all, Table S21 — the corpus median runs 32.8% to 38.4% against the declared 32.8%: 35.3% if the reference pipeline becomes target-encoded boosting, and 32.8% to 38.2% across the five instruments. The headline is not an artefact of which cell we called conventional.

And most cells are not cells whose sign the data determine. The median pair resolves 19 cells of the 180 it carries, and sign agreement between two draws from a distribution centred near zero is one half by construction, so a rate computed over unresolved cells is partly measuring that.

And a sign flip between two increments that both round to zero is not a misstatement anybody would act on. The rate counts a cell whenever its sign differs from the reference cell's, at any magnitude, and 8 of the 19 reference increments are themselves inside a hundredth of an AUC point. A *minimal practically important difference* is therefore declared — MPID = 0.01 AUC — and a disagreement is counted only when the reference increment and the cell's own increment both exceed it. The MPID is declared in AUC and Remark 1 forbids carrying it to another instrument, so the restriction is applied on the AUC sub-surface and the unrestricted AUC rate is reported beside it, in Table 7.

On the cells the corpus can resolve, the rate is 16.4%, and it

is not an AUC statement. Of the 890 cells whose whole-surface band excludes zero, 146 disagree in sign with the reference cell. Split by instrument that rate runs 0.0% on ROC AUC to 25.6% on Nagelkerke R^2: the pooled figure is carried by the instruments that are not the one the reference cell is read in, which is Remark 1 showing up in the corpus.

Imposing every correction this section's own argument asks for leaves 29 cells, of which the number that disagrees is 0. Restrict to the axes an analyst chooses, to the cells the band resolves, and to increments that clear the MPID at both ends, all on the AUC sub-surface. **The set that is being restricted is the inference family and not the admissible surface**: only 684 of the 5,808 admissible AUC cells carry bootstrap draws, and a cell with no band is neither resolved nor unresolved, so it can be neither kept nor dropped by the second restriction. Of those 684, 29 survive all three and the rate is 0.0%. **We therefore withdraw the claim that a one-number report misstates a sign at a rate worth quoting as a headline.** On the corpus's own AUC surface, once a sign has to be determined *and* worth acting on, disagreement is rare and the denominator is too small to carry a rate. Two corollaries follow and neither is comfortable for the conventional report. First, the reason so few cells remain is that *the data resolve so little*: 140 of the 684 AUC cells that carry a band resolve, 20.5% of them, which is the same story the whole inference surface tells at 890 of 3,420 across all 5 instruments; and 14 of the 19 pairs do not even resolve the cell a conventional paper would stand on. Second, **the surface still moves, and by a distance that dwarfs the MPID**: over the cells the band resolves, the mean distance from the reference increment is a median 0.0510 AUC per pair, and 0.0511 under the full restriction, both several multiples of the MPID. The one-number report is mostly not wrong about the *sign*; it is wrong about the *size*, and it does not say which.

6.5. Against specification-curve analysis as practised

The closest existing practice is specification-curve analysis [36]: enumerate the reasonable specifications, plot the curve, and test it against a sharp null by permutation. Table S22 runs it unchanged on *every* admitted pair — 19 of them, 100 permutations of a declared 24-cell sub-surface — beside the region label computed on the same cells, which makes "how often do the two verdicts part company" a measurement rather than an illustration. **That budget floors the permutation p-value at 0.010**, which is ample for the comparison made here — the test either rejects decisively or does not, and no conclusion turns on a p-value near the floor — but a reader wanting to quote a permutation p-value itself needs more permutations than this. Two budgets are declared, the sub-surface and a cap of 20,000 cases per pair that binds on 7 of them; Supplement S14 gives both.

They part company on 9 of the 19 pairs, close to half. The permutation test rejects the sharp null — the surface is decisively unlike one with no signal — and the same cells contain increments of *both* signs. A reader given only the test would conclude that the register matters and have no way to learn that whether it helps or hurts depends on choices nobody told them about. On 1 pairs neither resolves and the cheaper instrument suffices; the rest are mixed.

And a specification curve is a statement about the cells it enumerates. On 1 of the 19 pairs every cell of the sub-surface has one sign while the *full* surface is sign-changing or conditionally harmful. Nothing is wrong with either — they are claims about different sets — which is the point: without a declared denominator and a simultaneous band, a curve cannot distinguish "the sign holds" from "the sign holds on the cells I drew".

Neither subsumes the other. The permutation test answers *is there anything here*, costs one refit per permutation per cell and needs no band; the region answers *may I quote a sign*, costs a nested bootstrap, and is what a reader deciding whether to act needs. This paper's contribution is the second, and the first should be reported beside it.

6.6. Decision rules, in sample and out

The four rules of Section 4.7 are compared under the loss defined there, inside one instrument at a time. Out of sample — choosing on a random half of the admissible cells and paying on the other half — the one-number rule costs 0.00450 AUC of expected excess regret against 0.00011 for the majority rule and 0.00002 for the weighted mean. **Three results run against the surface's advocate and we report them**: the conservative rule is worst in 4 instruments of 5, the one-number rule's median excess regret is *zero* in every instrument, and under rolling-origin folds the ordering does not hold, at 0.00885 against 0.01922. We therefore do not conclude that a surface report is a better decision rule than a number; the case rests on the decomposition and the sign disagreement, and this is a check on those. Table S27 and Supplement S14.9 carry it.

7. Case study: a configuration management database

A configuration management database is the register an IT estate keeps of the items it contains, and it is bought partly on the argument that operational analytics need it. This section evaluates one against a named decision on a public log: will this case be reassigned to another group before it is resolved? The estate is a bank's and the log is BPI Challenge 2014 [12]; the register field is the affected configuration item, which takes 2,929 distinct values on this estate's cohort at incident creation. It is reported at greater length than any of the 19 log–target pairs of Section 6 because it is the case that motivated the study and the one on which the decision time — when in the service-desk process the prediction is made — decides the answer.

7.1. One log, two answers, and what separates them

This log appears twice in this paper. Section 6 applies the registered generic rules of Section 5, which admit 46,606 cases and define the target by the registered handover rule. This section uses the cohort and the reassignment target defined for the estate: 45,455 cases at prevalence 0.400 (0.372 in the test half), 31,818 in training and 13,637 in test. The estate's cohort is a strict subset of the registered one: the 1,151 cases the registered rules admit on top of it are those the estate's own cutoff excludes.

The two produce different answers about the register at the later decision time, and the difference is large enough to change the sign. Against intake plus group plus the knowledge reference, the estate's cohort and reassignment target give -0.0029 and the registered cohort and handover target give $+0.00768$, on which the planned contrast PC3's one-sided p-value rejects at the Holm-adjusted level while its interval does not exclude zero (Section 4.4). Table S11 takes the manuscript from the first to the second one factor at a time.

It is the target, not the cohort. Over the cohort $\times$ target design the target carries 99.3% of the variance in the increment, the cohort 0.0%, and their interaction 0.7%. Reassignment to another group before resolution has prevalence 0.400; handover between activity groups, under the registered rule, has prevalence 0.927. They are not the same question and the register is not worth the same to them. Two further candidate explanations were checked and neither is one: the two analyses reach the same attribute values on every row, so it is not a data-preparation difference; and they run the same pipeline at the same reference cell, so it is not a modelling difference.

decision time	baseline	n	prevalence	register levels	base auc	with auc	V	interval	resolved
T0 first touch, every call	intake	145478	0.146	4034	0.554	0.785	0.231	[+0.220, +0.262]	True
T1 first touch, escalating only	intake	44513	0.402	2445	0.552	0.754	0.201	[+0.193, +0.222]	True
T2 incident creation	intake	45455	0.400	2929	0.562	0.746	0.184	[+0.172, +0.204]	True
T2 incident creation	intake + group	45455	0.400	2929	0.644	0.748	0.103	[+0.095, +0.130]	True
T2 incident creation	intake + group + knowledge	45455	0.400	2929	0.805	0.802	-0.003	[-0.007, +0.002]	False
T2 incident creation, T1's population	intake	44513	0.402	2887	0.559	0.750	0.191	[+0.183, +0.213]	True

Table 8: The decision time implemented as an axis. Each moment carries its own population, its own register and its own admissible set; the PREDICTED TARGET is the same at all of them. τ_0 and τ_1 differ only in which cases are in scope. τ_1 and τ_2 differ in what is known AND in which cases are in scope, because 942 incidents have no interaction record; the sixth row is τ_2 restricted to τ_1's own population, and it is that row, not the third, that makes the τ_1-to-τ_2 step an information step alone. Every row is on the estate's own cohort and the reassignment target, under a declared tie-break, and every interval is the pointwise basic construction at 300 draws.

An axis nobody declares, and this paper was standing on it. Both analyses sort the log in time and split at seventy per cent of the row order, and 270 incidents share their timestamp with another; the sort this estate's analysis first used was not stable, so its tie order was neither declared nor reproducible, and it printed -0.0032 where the two declared rules both give -0.0029. That first number was never published and is not cited here as though it were: it is recomputed in the archive from that analysis's own loader (`scripts/s32_cohort.py`, `published_cell`), so the reconciliation below anchors on the object rather than on a recollection of it. Over 25 random tie-breaks the increment ranges over 0.00088 AUC without changing sign (0 of 25 draws positive), so no conclusion turns on it — but that range is 30.6% of the magnitude of the increment the declared rules give, produced by a choice no reporting standard asks anyone to state, which is why Supplement S16 now asks. Every number in this section is computed under a declared tie-break, by incident identifier, on the estate's cohort and reassignment target; the registered cohort's answers are in Section 6, among the 19 pairs, so the decision-time result below is a statement about a routing decision on this estate and not about handover in general.

7.2. Three decision times

A decision time fixes three things at once — which cases are in scope, what is known about them, and which fields are admissible — and the estate's surface is estimated at each of three moments.

The register's worth turns on which one the analyst occupies: at τ_0 it is +0.2310 [+0.2204, +0.2618] over 145,478 *interactions* — calls to the desk, of which only a minority ever become incidents, which is why that population is far larger than the 45,455 incidents the later moments carry — at τ_1 +0.2013 [+0.1930, +0.2224], and at τ_2 against the intake block plus the group it is +0.1034 [+0.0949, +0.1295] while against the block, the group and the knowledge reference it is −0.0029 [−0.0074, +0.0019] — not resolvably different from zero. **The step from τ_1 to τ_2 is an information step only on the matched population**: 942 of the 45,455 incidents have no interaction record, so the sixth row of Table 8 restricts τ_2 to τ_1's own cases and it is that row, not the third, that isolates the information. Supplement S14.15 gives each moment's population, register and admissible set.

7.3. The leakage tipping point, and register quality

Two further experiments on this estate are in Supplement S14.5: a leakage tipping point — the knowledge reference takes the register's increment to −0.0029, indistinguishable from zero, and reaches half its knowledge-free value at $\lambda = 0.64$, so an unprovable admissibility assumption is reported as a sensitivity analysis and not as a headline — and a register-quality axis under six declared mechanisms, three of them dependent, applied to both halves of the data with their parameters estimated from training alone (Supplement S11).

7.4. What this says to a configuration management programme

This subsection is the reading a service-management audience will want, stated in the estate's own vocabulary. It is a reading of *one* estate on *one* public log and it is offered as one; Section 10 says what a second would need.

Which field is which. The prediction is: *will this case be reassigned to another group before it is resolved?* The **register** f is the affected configuration item, 2,929 distinct values at incident creation on this estate's cohort and 3,019 on the registered cohort of Section 5, which is the cohort the layer ladder below is computed on. Its **coarsenings** are the item's type (13 values there) and subtype (65), each given its own role rather than swept into the baseline. The **free field** g is the assignment group — the routing queue, which is recorded without a configuration management database and is the field the register has to beat. The **intake block** B_0 is what the desk records when the case arrives: category, impact, urgency and priority. The **knowledge reference** is a pointer to a knowledge-base article, admitted as a further rung because whether it is available at the moment of decision is exactly the question Section 7.3 cannot settle.

The three moments, in service-desk terms. τ_0 is *an interaction is logged*: somebody has contacted the desk, every call is in scope, and 145,478 of them are. τ_1 is the same moment restricted to the calls that will escalate — the same information, a different population. τ_2 is *an incident record is created*: 45,455 cases, more known about each, and 942 of them with no interaction record at all. Table 8 is the ladder. **Read down it and the register's worth falls from +0.2310 AUC to +0.2013 to +0.1034 once the routing queue is in the baseline, and to −0.0029 — not resolvably different from zero — once the knowledge reference is admitted too.** A programme that justifies the register on a first-touch number and then deploys the model at incident creation is quoting the top of a ladder and standing on the bottom of it; Section 8.4 prices that in cases per thousand: 1.71 true positives promised per thousand cases against 0.003 delivered.

Which layer of the register pays. A programme does not choose whether to have a configuration management database; it chooses how much of one to populate and maintain, and that is a choice among *layers*. On the *registered* cohort of Section 5 and under its registered handover rule — 46,606 cases, one-hot logistic regression at the fixed temporal holdout, against the intake block at base AUC 0.641 — the item's *type* is worth +0.134 AUC over intake, its *subtype* +0.163, and the item itself +0.257; with the subtype already in the model the item still adds +0.109. **That ladder is not Table 8's cell and does not reconcile with it arithmetically**, because the table is on the estate's own cohort and its reassignment target — the pair Section 7.1 shows disagree about this register — so the increment over intake is +0.257 here and +0.1838 there and neither is the other one mismeasured (Supplement S4). **On this estate and the handover question the coarse layers are not free, and the finest layer is not redundant**: 52% of what the item is worth over the intake block is already reachable from a thirteen-value type field, and the item still adds +0.109 once the subtype is in the model. On the duration question the same ladder gives +0.045, +0.045 and +0.129, with +0.087 marginal, and the type reaches 35%. **So the split between the layers is a property of the target and not of the register**, and Supplement S4 carries two further logs on which every coarse layer is worth essentially nothing while the finest one is not. It is not a procurement conclusion: nothing here prices what a layer costs to populate or to keep accurate, and Section 10 says so.

What would have to change for the answer to move. Three things, in the order a programme could act on them. **A timestamped field history.** The knowledge reference decides this estate's answer and its admissibility is settled here by probability rather than by evidence; a register that records when each field value was written turns Section 7.3's tipping-point analysis into a measurement. **A complete population at the decision time.** 942 of the incidents have no interaction record, and the τ_1-to-τ_2 step is an information step only on the cases that have both; a desk that can identify its own cases across the two systems recovers that comparison. **A target the desk actually acts on.** The register's sign changes with the target on this log (Section 7.1), and reassignment is the question this desk routes on; a programme evaluating a register against a generic handover rule is answering a question nobody asked it.

7.5. The absorption, and why the ratio is secondary

The quantity a report of this kind usually leads with is the *relative* reduction R: the share of the register's apparent value that a single free field absorbs. The absolute absorption $D = V(f \mid B_0) - V(f \mid B_1)$ is reported first, with its own interval, and R second with a Fieller set and the set's kind. On this estate at the reference cell D is +0.0804 [+0.0626, +0.0914] and R is 0.437 with a bounded Fieller set. This ladder is the favourable case: 0 of the 30 reductions it produces have a Fieller set that is not a bounded interval, because the denominator is resolvably positive at every rung of it. Across the corpus that is not so — 5,282 of 10,584 reductions have a non-bounded set, and a percentile interval reports a bounded one for every one of them — which is why the ratio is secondary everywhere and not only here. The full ladder is Table S33.

8. Decision-analytic evaluation

An increment in AUC is not a quantity a service desk can act on. This section reads the same surface in net benefit, which is: the number of correctly nominated cases per thousand arrivals, net of the false nominations, at a declared exchange rate between the two.

8.1. The operating point is a cost ratio, not a knob

At threshold θ the net benefit of a rule that nominates when $\hat{p} \geq \theta$ is

$$\mathrm{NB}(\theta) \;=\; \frac{\mathrm{TP}(\theta)}{n} \;-\; \frac{\mathrm{FP}(\theta)}{n} \cdot \frac{\theta}{1-\theta},$$

so $(1-\theta)/\theta$ is the number of false nominations the desk is willing to accept for one true one. Choosing θ is choosing an operational cost ratio, and it belongs to the analyst's declaration rather than to a grid search. We report the whole grid — 31 points from 0.05 to 0.80 — and name three points a desk might occupy: $\theta = 0.10$, or 9.0 false nominations accepted per true one, where a review is cheap and the desk nominates liberally; 0.30, or 2.3, where a review and a miss cost about the same; and 0.60, or 0.7, where review capacity is scarce. A reader whose desk sits at a ratio not in that list reads their own point off Figure S3: the operating point is the reader's to choose, so the analyst reports the curve rather than a point on it.

8.2. Calibration comes first, or the threshold means nothing

A threshold is a statement about probability, so a model whose probabilities are wrong is being read at a threshold it does not have. Every decision curve here is therefore computed on *calibrated* scores, with the calibrator fitted inside the training half and rebuilt inside every bootstrap refit, and a registered rule excludes an arm whose calibration slope leaves $[0.5, 2]$.

The window is a declared convention and not a derived bound, and the exclusion is deliberately not repaired by recalibrating the failing arm: recalibration is already a level of the pipeline axis, so repairing an arm with it would replace one specification by another and report the result under the first one's name. Supplement S14.6 gives the argument, the rule, what it removes and what the curves look like without it.

8.3. Simultaneous bands change what can be said

The decision-curve family is declared separately from the scalar surface and carries its own max-t critical value, because net benefit and a scalar instrument are not the same scale. That critical value does not reach the level it is named for, and Section 9.4 measures by how much on families matched to this one in size, draw count, cross-cell correlation and tail. The widening it measures is applied here rather than reported: the operative critical value below is 9.33, not the 3.57 the multiplier bootstrap returns.

Across the 31 operating points of this estate's curve **the pointwise interval declares the increment resolvably positive at 24 of them and the widened simultaneous band at 5**, every one of the second at or above $\theta = 0.68$; the harmful counts are 0 against 0. All are built from the same draws, so what differs is the family and the level it is bought at: those points are the price of a statement about a *range* of thresholds rather than about each one separately, and a paper that reads a decision curve across its range while pricing it pointwise has claimed the first and paid for the second.

The widening costs 13 of the 31 operating points, and the uncorrected count is printed beside the corrected one rather than dropped: the same draws declared 18 before the widening, and the median band width goes from 0.0136 to 0.0354 against the pointwise interval's 0.0071. The factor applied is 2.61, the upper end of the range Section 9.4 measures on families that carry this corpus's tails and no degenerate cells; the lower end 2.20 returns the same count, so which end is taken is not carrying the result. The choice between the two kinds of matched family does move it, and the alternative is therefore printed rather than argued away: the all-cells factor 2.58, measured where the matched families are built to carry cells at which both arms treat every case alike — 638 of them across the corpus's curve families — removes a further point, to 5. The admissible factor is the operative one because this family is not the degenerate case — 2 of its 248 cells are zero in at least nine draws of ten, against a corpus share that reaches 14.3% — and the all-cells count is printed beside it because that argument is a judgement rather than a measurement.

The pointwise interval is not widened by a family-wise factor; what it covers is a separate measurement and Section 9.4 reports it. What the decision curve still does not receive is the (n, K) calibration of Section 6.3, which is estimated for the scalar family and is fitted to restore pointwise rather than family-wise coverage. Supplement S14.7 carries the comparison and Figure S3; Table S25 prints the curve.

8.4. The decision, in units a service desk uses

The comparison is run in net benefit per thousand cases, over a declared distribution of exchange rates a desk plausibly holds (0.08 to 0.36, centred 0.19), and only on models that pass the registered calibration rule — which excludes 40 of 118 arms and 3 of 19 pairs. **That count is itself the strongest operational finding here**: a decision curve read off a model whose probabilities are wrong is read at a threshold the model does not have.

And the rules separate on most pairs, with the excess falling on the conservative rule rather than the one-number rule. The one-number rule's median excess is 0.000 per thousand against the conservative rule's 0.048, and the cross-pair means are 0.056 and 6.092: what the medians hide is a tail, and it is the conservative rule's. The four rules do not all lose the same amount on 11 of the 16 pairs the calibration rule leaves, and all four are identical on the other 5. Where they do separate, the one-number rule is strictly the worst on 3 and tied for worst on 2, at a maximum of 0.412 per thousand. **The case against one-number reporting does not rest on this design.** Larger than anything the choice of collapse rule costs a one-number report is the decision time: the register promises 1.71 per thousand at the earlier moment and delivers 0.003 at the later one, a shortfall of 1.70. Supplement S14.10 carries the rules, the exchange-rate distribution and the exclusions; Tables S23 and S27 the numbers.

9. Simulation against a known answer

Everything so far has been measured on real logs, where the answer is not known. This section measures the estimator where it is. The full tables are in Supplement S15 and the diagnosis of the one world that failed is in Supplement S6.

9.1. Six worlds, two estimands

The simulation uses six worlds — four of them misspecified for the estimator — at 1,000 replicates each, with the covariate space finite so that the population AUC of any scoring rule is computed in closed form rather than simulated. Two estimands are separated because under misspecification they differ: V_{oracle}, the increment between the two Bayes-optimal rules, and V_{limit}, the increment between the two fitted pipelines' limits. A resampling interval can only cover the second, and reporting its coverage of the first as the estimator's coverage is a category error. Supplement S6.1 describes the worlds and the marginalisation a corrupted register forces; Table S37 reports both estimands.

9.2. Coverage, bias and width

Coverage estimates carry a Monte Carlo standard error of about 0.7 percentage points at 1,000 replicates, which is the resolution every comparison here should be read at; Figure S5 draws them by construction and world.

Where the refit helps, and where the percentile interval fails. The nested basic interval covers a median 93.0% against the fixed-model interval's 92.1%; on the sparse world the *percentile* interval collapses to 37.2% and the basic construction restores it to 89.9%, which is why every interval and band here is the basic one (Supplement S6.2; Table S37).

What no interval repairs. Against V_{oracle} rather than V_{limit}, every construction undercovers on the drifting world — 65.8% at best — and it should: when the coefficients move between eras the estimator is not consistent for the oracle quantity, and the largest gap is 0.0161 in AUC units. Under misspecification an interval around the estimator's own limit is not an interval around the quantity a reader cares about, and no resampling scheme repairs that.

9.3. Sample size, register cardinality and block length

A construction that works at one sample size, one register cardinality and one block length has been validated at a point, not in a regime. Sample size and register cardinality are therefore varied *jointly*, up to $K = 3{,}019$ levels — the cardinality the case study's log carries on the *registered* cohort of Section 5, which is the cohort the corpus admits it under; the estate's own cohort in Section 7 carries a different one at each of its three decision times (Table 8) — and the block length is varied around its rule of thumb.

The ratio K/n captures most of the dependence, and not all of it. Across the plane the basic interval covers a median 83.3% and a minimum 21.3%, the last at a ratio of 0.377, and it is short of nominal everywhere. It is not sufficient: two cells at nearly the same ratio cover 0.780 and 0.613, a gap of 3.2 combined Monte Carlo standard errors, and holding the register fixed while moving the training half from 2,000 rows to 8,000 moves coverage from 0.453 to 0.613. **This corpus sits in the regime where that matters**: its median pair has $K/n = 0.106$ and 10 of 19 pairs are above a tenth, so a nominal band on them is *narrower* than a 95% interval should be. Supplement S9 derives a calibration against the plane in full, and Section 9.4 reports the measurement that keeps it from being applied.

Where the interval stops containing its own estimate. The denser plane the calibration of Supplement S9 is fitted on — a different and larger object from the 10-cell grid above, at 500 to 180,000 training rows — locates the crossing Section 4.1 describes. At a fixed ratio of 0.125 the basic interval straddles its own point estimate in every replicate up to 8,000 training rows and in only 1.6% of them by 50,000, because the bootstrap displacement is a bias that does not shrink with n while the half-width does. Past that crossing the inflation factor is not defined — it is a widening about the estimate, and there is nothing to widen about — and 4 of the plane's 33 cells have none for that reason, every one of them at 50,000 rows or more. Section 6.3 reports what that leaves the corpus, and Supplement S9 describes that plane's design.

Block length. 3 lengths are examined — half the rule $\lceil n^{1/3} \rceil$, the rule, and twice it. Coverage moves with the length, spanning 3.6% within a world, of the order of two standard errors; but the comparison the paper rests on is unchanged at every length, the basic interval beating the percentile interval on the sparse register by at least 46.4% at all 3.

9.4. The band's family-wise coverage

Everything above validates a *pointwise* interval. Two of this paper's three reporting objects are functions of a *simultaneous* band, and until this section nothing measured what that band covers.

The design, and why its answer is an upper bound. The truth is zero at every cell of a synthetic family. One draw from the sampling distribution gives the point estimate, and as many further draws *from that same distribution* as the corpus family it is matched to carries give the bootstrap — so the bootstrap is not an approximation of the sampling distribution but is it. Every way a resampling scheme can be wrong — dependence it fails to reproduce, a refit that comes apart, a block length misjudged — is switched off, and what is left is the object under test: a standard error and a critical value both estimated from the same handful of draws. A coverage measured here is therefore the most the real procedure can attain, and a shortfall here is one that no improvement to the resampling repairs. The families are matched to this corpus in size, draw count, cross-cell correlation, effective rank and per-cell excess kurtosis; the matching is fitted by grid search against the corpus's own draw files and reported beside its target rather than asserted. Coverage is over 2,000 replicates in each of 24 cells, so no coverage here is determined worse than 1.1%.

Three regimes, because "the draws are heavy-tailed" turned out to name two different things. Under *Gaussian* draws the multiplier's family-wise coverage is a median 87.6% and a minimum 75.8%. Under draws carrying this corpus's tails, whose median cell has an excess kurtosis of 0.52 on the surface families and whose ninetieth percentile is 7.9, the family matched to the reported design covers 91.6% and the 7 surface families run 80.3% to 91.6% as the cell count and the draw count move around it; the decision-curve families cover a median 81.5% and a minimum 74.1%, and are reported apart from the surface ones throughout, because the widening each needs is different. Under the third regime, which adds the degenerate cells described below, they are 82.6% and 43.8%.

What is wrong with the decision-curve families is not a tail. At a threshold where both arms treat every case alike the net-benefit difference is zero by arithmetic, and 638 cells of the corpus's curve families are exactly zero in at least nine draws of ten, against 0 anywhere in the surface families. They are what makes q_{emp} large there; excluding them moves the median ratio q_{emp}/q from 2.68 to 2.24 and does not dissolve the disagreement. Supplement S9.6 carries the mechanism and Table S40 the regime with and without them.

How far short, in the units a widening works in. A critical value is widened multiplicatively, so the useful summary of the shortfall is the factor that would have closed it. On the surface families that factor is 1.36 to 1.88; on the decision-curve families, 2.20 to 2.61. **No widening is applied to either.** The (n, K) calibration Supplement S9 derives was fitted to restore *pointwise* coverage on a plane built around a different inference surface; at the median pair it supplies 1.34, which is below the bottom of the surface range above and well below the decision-curve one. Applying it would have widened these bands by less than the measurement says they need, while presenting them as corrected — a worse position than leaving them nominal, because it spends the resolution and buys a label the coverage does not support. The labels this paper prints are therefore nominal and anti-conservative by the factors quoted here, which is a defect this paper has measured and has not removed. Section 10 says what follows.

The draw count is *not* the binding constraint, which is the opposite of what the asymptotics suggest. At the modal surface family the multiplier band covers 81.2% at 33 draws, 91.2% at 150, 91.6% at 400 and 91.3% at 1,000 — it rises steeply out of the smallest budgets and then **plateaus short of its level**, so a corpus that can afford more draws does not thereby buy coverage. This corpus runs at 400 throughout, on the plateau and not below it. The per-cell interval underneath the band moves 90.3% to 94.6% over the same range: part of what the band loses it loses before any critical value is chosen, and Section 9.2 validated that construction at a refit count this corpus does not reach on most pairs. No estimator recovers the difference. The best of the five, the empirical quantile at the upper end of its own order-statistic interval, reaches 91.6% at 1.83 times the width and still falls to 88.8% at the smallest family; a Rademacher wild multiplier, which preserves the observed draws' magnitudes instead of replacing them by Gaussians and is the natural repair for a tail problem, is worse than the Gaussian multiplier at 78.5%.

What the simulation does not establish. It certifies the pointwise constructions on six worlds, a plane of 10 (n, K) cells and 3 block lengths, and the band on the 24 matched families of Section 9.4 — on nothing else. The stationarity and mixing the moving-block bootstrap needs are assumed and not tested; the split point is fixed within a draw, so split-position uncertainty is outside every interval reported here; and the drifting world shows what happens when the estimator is not consistent for the quantity a reader cares about.

10. Threats to validity

Organised as threats rather than as a list. No result is computed here: every number this section quotes is the same macro, from the same result file, as the section that establishes it, so the two cannot disagree — but many are quoted twice, and a reader who wants the argument rather than the summary should read them where they are established.

Construct — this is predictive performance, not value. Nothing here measures what a register costs to acquire or maintain, whether acting on a prediction changes an outcome, or what a data programme is worth; that includes Section 7.4's layer comparison, which prices no layer's upkeep. The corpus is also a family of related questions rather than replications — the two registered targets label the same case identically on a median 51% of cases, and the registers run from a configuration item to a legal article — so every corpus statistic is about *sensitivity* and every one is reported on the 8 ITSM pairs as well (Table S17); Table S13 names each pair's own construct threat. The registered handover target on the case study's log is close to degenerate at prevalence 0.927 and carries more of the corpus's admissible cells than any other pair (Section 5), and Section 7.1 shows the register's sign turns exactly there. Whether a generic handover rule is a meaningful target on ITSM logs, where reassignment is what a desk acts on, and whether the prevalence window should close below 0.95, are open; we did not narrow the rule retrospectively, because doing so after seeing which pair it admits is what a registered protocol exists to prevent. The quality mechanisms are applied to both halves with their parameters estimated from training alone (Section 7.3), so they are the operational counterfactual and not a train–test mismatch — but they remain *constructed*, and validating them against a genuinely incomplete register is the obvious next study.

Internal — the band does not attain its nominal level. This is the sharpest limitation in the paper. Section 9.4 measures family-wise coverage against a known answer over matched families whose bootstrap is *ideal*, so every coverage there is an upper bound on what the procedure attains, and on the family matched to the design this paper reports it is 91.6% against a nominal 95% — short by more than five Monte Carlo standard errors, so the shortfall is not a resolution artefact. The safeguard we called conservative moves coverage by less than a point, because it controls the Monte Carlo error in q and not the error in what q estimates. More draws were the repair this suggested, and the paragraph below reports what they bought, which is less than the asymptotics promise. **Every region label and every ρ here is therefore a descriptive diagnostic and not a guarantee**, which the front matter now says.
The two families are not equally exposed and are no longer corrected alike. Section 9.4 gives both shortfalls and Section 8.3 what applying one to the decision-curve families cost. **The correction is narrower than it looks**: it is fitted where the bootstrap is ideal, so it closes the shortfall this design can measure and not whatever the real resampling adds to it; and it leaves the surface families with no widening at all, because the only one this paper has fitted is smaller than they need. **Their** region labels and ρ are the conclusions to distrust.

Internal — two repairs this paper named, both run, and neither did what it was named for. Two changes this paper's own analysis identified as the first to make are run here, and what they bought is reported. **Both retire a mechanism this paper asserted, and neither fixes the thing it was asserted about.**
Level loss does not explain the displacement. A multinomial resample holds about $1 - e^{-1}$ of a high-cardinality register's levels, and that was offered as the reason the bootstrap distribution is displaced. Under weights every row has positive weight in every draw and the level share is 100% against 80.8% — the mechanism is gone by construction — and the displacement's median moves only from 0.0050 to 0.0041 (Section 4.1). What the change does remove is a defect the mechanism genuinely caused: cells a draw drops entirely, which get no band at all, and which were counted as *unresolved* beside cells that were banded and straddled zero. There were 150 of those on the previous surface, 270 under the multinomial scheme on this one, and there are now 0.

More draws do not fix the coverage. The binding constraint on the band's family-wise coverage is not the draw count, which the construction's asymptotics would suggest and this design's own measurement refutes. On families matched to the surface this paper now reports, coverage is 91.2% at 150 draws, 91.6% at 400 and 91.3% at 1,000: **it plateaus**. This corpus runs at 400 draws throughout and the band still does not attain its level — 91.6% on the family matched to that design under this corpus's tails, and 84.3% under a non-zero truth, against a nominal 95%. The widening that would close it is 1.36 to 1.88 on the surface families, which is *larger* than the 1.18 to 1.77 the only calibration this paper has fitted would supply — which is why that calibration is reported and not applied.

So every region label and every ρ in this paper remains a descriptive diagnostic and not a guarantee, which the front matter says, and the conclusions that rest on one are the ones to distrust. Two things follow that are worth more than the repairs would have been. The first is that a mechanism can be removed completely and leave its supposed consequence standing, which is a caution about every explanation of a resampling artefact offered without the counterfactual — including the two offered here. The second is that this is the paper's own thesis arriving at the paper: a claim about why a number moves is a claim about a design, and it does not survive being asserted rather than run.

Internal — further assumptions, named, and one of them measured. The split point is fixed within a draw, so split-position uncertainty is outside every interval here. **A second sort is undeclared, and unlike the one Section 7.1 reports it is not repaired here**: the register-quality mechanisms cut a cumulative-count curve inside a group of tied identities, so *which* identities a degradation removes rests on an order nothing pins. The affected quantity is the quality axis and not the increment at the reference cell, and repairing it moves every number on that axis — which is why it is measured and recorded rather than corrected in a round that is not regenerating them. It is the second instance of one defect, which is why Supplement S16's tie-break item is stated over every sort a procedure cuts on rather than over the place it was first found. The moving-block construction needs stationarity and mixing that an event log is not guaranteed to have, and the same assumption is what makes Definition 2's population limit well defined under a temporal split. **Stationarity is now measured and it does not hold everywhere**: cutting each pair's test half into 5 consecutive blocks and reading the observed spread of the increment against a null that re-partitions the same rows at random, 6 of 19 pairs sit in the upper tail of that null at the declared level $\alpha = 0.05$, and 11 carry both signs between blocks of one test half (Supplement S13). Only 1 pair shows a monotone trend, so this is heterogeneity and not decay. **On those pairs the interval and the band understate what a reader would see on a different stretch of the same log**, by an amount this design cannot sign; time is not a declared axis of $\mathcal{S}$ and adding one is the change the measurement implies. Mixing is still not tested. The band's family-wise coverage is asymptotic, as is the multiplier bootstrap estimating its critical value, and the block length is a convention, $\lceil n^{1/3} \rceil$, which moves coverage by up to 3.6% within a world. The coverage calibration is itself an assumption — it corrects the studentised statistic's *scale* and not the shape of the maximum's distribution, it is fitted pointwise and applied simultaneously, and it is measurably misspecified ($\log n$ enters at $t = 8.3$) — so a reader who rejects the transfer should rely on the objects that need no band: the point estimates, the decomposition, the disagreement rates and the regret comparison.

Conclusion — two claims a specification surface invites and this evidence does not support. Section 11 states what the decomposition and the disagreement rate leave standing once the pooled surface's resampling is netted out and the three restrictions are imposed; both readings a pooled surface invites are withdrawn there rather than here. Four qualifications belong to this section because they are limits and not results. Both decompositions are reported under all three declared measures because there is no neutral measure — across the three the higher-order share moves by 24.3% — and the within-instrument decomposition is primary because the decomposition inherits the metric's scale (Proposition S1). Region labels remain relative to an inference family of a median 16.7% of a pair's admissible cells, now under a minimum resolved share that withdraws the direction on 5 of 12 pairs, and they are not partitioned by axis kind, so they stay conditional on the quality axis being averaged over. And specification regret reports a null and is retained as a robustness check rather than a contribution.

External — the design space is declared, not complete. $\mathcal{S}$ has nine axes in Definition 1 and the per-pair factorial varies 5 of them (Table 3); the decision time, the operating point and the cohort vary on the case study's log alone, and the target indexes surfaces rather than varying within one. It could have more — tuning protocol, missing-data treatment, target horizon, label-noise model — so the honest claim is “the increment moves this much over *this* space”. **The structural omission is that prefix length, prefix bucketing and sequence encoding are not axes here and cannot be**, because this study predicts at one prediction point from creation-time attributes, and those are the axes predictive process monitoring varies most. **We therefore claim no result about that literature’s own benchmark**: what is reported is case-level outcome prediction from event-log attributes. A prefix axis is nonetheless run on one log, because the obvious question is whether this paper’s central claim is a property of that literature’s axis too (Supplement S12). It is. Holding the register and the estate fixed and moving only the prefix, the increment falls from +0.257 AUC at case creation to +0.064 two events later, does not decline monotonically beyond that, and is undetermined in sign at 2 of 6 prefixes. **Case creation — the point every other result here occupies — is the most favourable prefix for a creation-time register**, and two studies of this field reporting from the two ends of that axis would differ by a factor of 3.7 in point estimates without either being wrong. What this measurement supports is the fall away from case creation and not a shape for the rest of the axis: one log, one register and no sequence encoder is a pilot, and the structural omission stands.

External — three narrower limits, named. The *inference surface* is smaller than the declared one and always will be, because a draw refits every arm of every cell (Section 4.2), so every region label is a claim about a fraction of the space the point estimates cover — and a smaller set is a set on which *uniformly beneficial* is easier to earn, which Section 6.3 reports the one label that turns on. The *pipeline axis* is now crossed on the 8 ITSM pairs and not only on the case study’s log, which narrows this limitation rather than removing it: the pairs outside that family still carry family and encoding confounded, and what the wider crossing found is not a clean replication (Section 6.2), so the case study’s ordering should be read as the common case and not as a corpus fact. And the *case study* is one bank’s estate on one public log, with the availability question at its centre settled by probability rather than by a timestamped field-value history; Section 7.4 names the three changes that would move which answer. Finally, we do not measure reporting practice: the pilot that attempted it is in the archive, and it is a hypothesis rather than evidence.

11. Conclusion

The incremental predictive performance of a recorded field is a functional of design choices, not a number. This paper made that operational: a declared design space, a surface over it, an exact decomposition of the surface's variance that separates the choices an analyst makes from the resampling the design also contains, simultaneous bands whose coverage is measured rather than assumed, and region labels that say where a sign is determined and withhold a direction where too little of the surface resolves to support one.

On every pair, at least one admissible specification makes the register worth less than nothing. That is a statement about point estimates and needs no band. It is also nearly guaranteed by a space that includes three-quarters-deleted registers, so the version worth carrying is the restricted one: **on 7 of 19 pairs the baseline alone — among the baselines an analyst would actually build — moves the increment further than the increment itself**. Four further results are worth carrying away.

Most of a pooled specification surface is resampling, and net of it different pairs turn on different analyst choices. The components involving the split carry a median 71.7% of the variance. Net of them the largest first-order index at the median pair is 38.8% against a higher-order share of 26.3%: the interactions carry more than any single axis on 7 of 19 pairs, against 13 before the resampling stratum is removed, so *most of the variance belongs to no single axis* is not a claim this evidence supports (Section 6.2). What it does support is that which axis leads is a property of the pair — the baseline rung on 8 pairs, the register-quality condition on 5 and the pipeline on 6. An analyst therefore cannot learn from somebody else's paper which of their own choices their answer turns on — but neither should they read a pooled specification curve as evidence that everything matters.

A one-number report is usually right about the sign and usually silent about the size. Once a sign has to be determined by the data *and* worth acting on at 0.01 AUC, the number of the 29 *AUC* cells surviving all three restrictions that contradict the conventional report is 0 — the MPID is declared in AUC and Remark 1 forbids carrying it further. What the conventional report does not say is that the cells it does not stand on lie a median 0.0510 AUC away from the one it does, and that on 14 of 19 pairs the data do not determine the sign of its own cell at all.

The decision time is an argument of the estimand, not a preliminary. It fixes which cases are in scope, what is known about them, and which fields are admissible. On the case study, moving the prediction from first touch to incident creation costs -0.0297 AUC through the population, a further -0.0099 through the information once the population is held fixed, and -0.0076 more through the incidents that have no interaction record at all — before any baseline changes; and the same log's answer changes sign when the target changes, with 99.3% of the difference on the target and 0.0% on the cohort. A desk that adopted this register on its *first-touch* number and deployed it at incident creation would be promised 1.71 true positives per thousand cases and receive 0.003.

And a threshold is not always a cost ratio. On 40 of 118 fitted models the calibration slope falls outside $[0.5, 2]$, so the threshold at which a decision curve is read cannot be interpreted as the cost ratio the decision-analytic argument requires (Section 8.2).

The practical recommendation is narrow and cheap once the surface exists: report the surface, the decomposition with the resampling stratum separated, the region and ρ with the triple they rest on; state the decision time and the target as parts of the estimand; declare a minimal practically important difference before reporting any disagreement rate; make the absolute increments primary; and use a ratio last, with a Fieller set, and only where its denominator is resolvably positive.

CRediT authorship contribution statement

Sudhir Dixit: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

Declaration of competing interest

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Anthropic Claude (`claude-opus-5`, 2026-08-24) in order to draft and edit prose and to write analysis and verification code. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.
This is a use beyond the language-and-readability editing an author may make without comment, which is why it is declared, and the two halves of it are separated here because the policy this section answers is about the first: the prose was drafted and edited with the tool, and the analysis and verification code was written with it, which is a use in the research process and is disclosed rather than assumed to be covered. The tool is named with its version and the date it was used, because a disclosure that names a product and not a version is not reproducible; every session's model identifier and date are in `AI-USE.md` in the archive. **No generative model produced, imputed, augmented or selected any datum, any result or any citation behind a claim of this paper**: every result is produced by public code that `REPRODUCE.md` re-runs by one command, and the reference list was checked against the published records. The one place where a machine-assisted judgement enters a study is the literature pilot of Supplement S16, which is declared where it is described, and no claim in this paper rests on it.

Data availability

All data are public. Event logs are obtained by DOI: BPI Challenge 2014 (https://doi.org/10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35), BPI Challenge 2013 incidents (https://doi.org/10.4121/uuid:500573e6-accc-4b0c-9576-aa5468b10cee), and the further public logs whose DOIs and SHA-256 checksums are listed in `data/corpus/CHECKSUMS.txt` — 26 files are downloaded and checksummed in all, of which the registered role-assignment rules of Section 5 admit 13, yielding the 19 log–target pairs analysed here. The counts are different quantities and Table S12 accounts for every downloaded log that is not among the admitted pairs, in one of two ways: with the registered exclusion code the rules gave it, or with the code HELD_OUT, which marks the held-out set — 1 of the checksummed files, downloaded for the out-of-corpus test and never offered to the admission rules. The literature pilot's frame, sample, codes, adjudications and evidence dossiers are in the archive; the retrieved full texts are other publishers' copyrighted PDFs and are not redistributed, but `scripts/s06_audit2.py --fetch` re-retrieves them from the DOIs in `results/s06_sample.csv`. All derived results (`results/*.csv`), all figures and all code are in the archive below.

Code availability

Code, derived results, the pre-registered protocols, the amendment log and the verification harness are archived at the archived release cited in the data-availability statement (DOI reserved, inserted at proof) (release v27.0) and developed at https://github.com/sudhirdixit1/cmdb-routing-queue-baseline. The archive contains a `Dockerfile` and a `requirements.lock` pinning every transitive dependency; `REPRODUCE.md` gives the exact commands and the expected runtime. **Bit-exact reproduction is claimed inside that container — whose image digest the archive records at release, the `Dockerfile` in the meantime pinning by tag — and not outside it.** On a different processor architecture, cells whose design carries no high-cardinality register agree to floating-point tolerance, while cells that carry one can differ materially in every metric — the register ties test rows to equal predictions, and a difference in the last bit unties them, so a rank-based metric steps. The divergence tracks the register's cardinality and not the learner: it is present in both model families and absent from the low-cardinality rungs of each. `REPRODUCE.md` measures it for both families and states a tolerance per condition, and `scripts/check_reproduction.py` re-runs part of the surface and fails if one is exceeded. A reader checking a *digit* therefore needs the container; a reader checking a *conclusion* does not.

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